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Master Thesis

RSS-based localization: using distributed spectrum sensing for 5G networks

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Declaration

I declare that the work is entirely my own and was produced with no assistance from third parties.

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Erlangen, October 2, 2023

Abstract

One purpose of wireless sensor networks (WSNs) is to observe environmental events and objects with minimal human intervention. This could be achieved by deploying numerous low-cost sensor nodes that relay measured data to the central server. The number of nodes required depends on the application, but it is typically quite large, for example, in the case of Internet of Things (IoT). Accurately locating a transmitter node is crucial for answering important questions about events or objects reported by the network, for instance, spectrum occupancy, transmitter, interferers present in the network, and location services used by end users. The received signal strength (RSS) attenuation with distance is an effective method for estimating the location of both stationary and mobile wireless objects. This is possible because the received signal strength indicator (RSSI) is a standard feature in most sensor node radios, requiring no additional hardware.

The motivation behind the work has been arising from literature research based on the importance of active transmitter position in radio environment map (REM) construction. REM enables the radio environment database to characterize the propagation environment for a communication network, estimating characteristics such as frequency channels in operation, interference, active transmitter's accurate locations, wireless channel estimation, and modulation schemes used in communications. The main contributions in the thesis are evaluating RSS-based active transmitter localization using MSE and LiVE-based methods in a static scenario employing the use of log-normal shadowing and correlated shadowing to characterize the occupied spectrum (frequency in operation and position of the transmitter) using a dedicated sensor network. The evaluation of the two localization methods helps us to choose the MSE method as a favorable method in terms of trade-off between localization accuracy and faster simulation time in MATLAB. The thesis also evaluates the use of orthogonal division multiplexing (OFDM) waveform as an alternative transmitter signal (using signal processing) instead of using static transmit power, as is used widely in literature,

The current literature research and reviews propose RSS-based localization or REM construction methods, while this work implements both localization and REM construction. We employ the results of the MSE-based method to estimate a REM in the case of additive noise and an ideal scenario without noise to infer the presence of an active transmitter in an area of interest. Additionally, the literature review outcomes explain why REM as a technology is not only limited to cognitive radio (CR) but is also advantageous for various WSN applications.

Keywords: Wireless communication, Distributed spectrum sensing (DSS), Radio environment map (REM), Received signal strength (RSS), 5G, Wireless sensor network (WSN).

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List of Abbreviations

LTE	Long Term Evolution
RSS	Received Signal Strength
REM	Radio Environment Map
IoT	Internet of Things
ToA	Time of Arrival
AoA	Angle of Arrival
TDoA	Time Difference of Arrival
GMM	Gaussian Mixture Models
ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machine
URLLC	Ultra Low Latency and Reliability
WSN	Wireless Sensor Networks
UAV	Unmanned Aerial Vehicle
UWB	Ultra Wide Bands
PRS	Positioning Reference Signal

SRS	Sounding Reference Signal
DNN	Deep Neural Network
V2X	Vehicle to Anything
PSD	Power Spectral Density
QAM	Quadrature Amplitude Modulation
SRCKF	Square Root Cubature Kalman Filter
DEKF	Diffusion Extended Kalman Filter
LoS	Line of Sight
ITS	Intelligent Transportation Systems
CP	Cyclic Prefix
MIMO	Multiple Input Multiple Output
CV2X	Cellular to Anything
CS	Compressed Sensing
BS	Base Station
AP	Access Point
HST	High Speed Trains
MAP	Maximum a Posteriori estimator
CBRS	Citizen Broadband Radio Service

1 Introduction

1.1 Background and motivation

In the past two decades, we have witnessed the immense growth of cellular and wireless local area networks, starting from 3G to 5G New Radio (NR). This growth essentially means technological advancement in services, seamless connectivity, reliability of connection and data collection, accessibility, portability (size of mobile handsets), and location-based services. Every end user is entitled to access information in a timely and accurate manner. An important criterion to gain access to information is location-based services [1]. Most applications on mobile phones or experimental research studies (on robot positioning, satellite-based navigation) need to provide the user's position to gain access to the internet or use application-based services on their devices. This brings the need for localization of end-user not only for the general public in the sense of using mobile phones but also its importance in the fields of robotics, indoor automated warehouse management, autonomous vehicles, and drone-based applications, to name a few. Localization of an active transmitter helps to characterize the radio environment and the spectrum in use to avoid interference and maintain seamless connectivity. Localization problem found its root through the global positioning system (GPS) in satellite-based services for outdoor localization; however, the GPS signal finds its shortcomings in the urban and indoor environments as they are heavily affected by multipath, fading loss, and interference degradation [2], [3], [4]. As a result, it's essential to look for technology that is independent of the environment and easily accessible to avoid setting up a separate infrastructure for localization. Figure 1.1 describes the use of various technologies for localization or positioning applications in indoors, outdoors, or rural areas against the resolution or accuracy of localization.

This idea manifests itself into a solution that can be accessed using cellular and wireless local or wide area networks employing the use of RSS measurements or power measurements from these technologies [3]. RSS measurements are readily available and easy to process, as will be explained in detail in the following chapters. RSS-based transmitter localization has found its importance in many literature research [6], [7], [8] as it provides added advantages to the GPS-based positioning in terms of area served, very low cost and low power consuming hardware. Since deploying 5G NR and its diverse use cases serving many IoT, indoor localization, high mobility, and low latency applications, it has become essential and of high priority for network operators to characterize the radio environment and sense radio propagation channels. This characterization of the radio environment would enable the network to be highly intelligent and self-organizing, meaning to allocate spectrum, time slots, and available channels effectively without wasting bandwidth and compromising the allocated users' connectivity and data rate. REM is a highly intelligent enabling technology that provides us with the network cognition we desire in such a diverse and complex application infrastructure. Figure 1.2 describes the exact

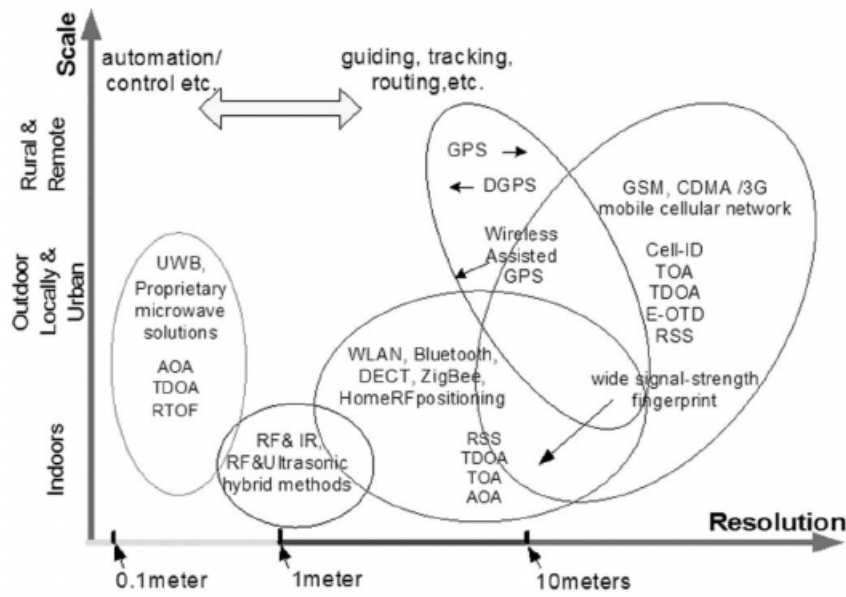


Figure 1.1: Wireless technologies used in localization techniques [5].

simulation environment for RSS-based localization using measurements taken from a dedicated sensor network and REM or power map construction.

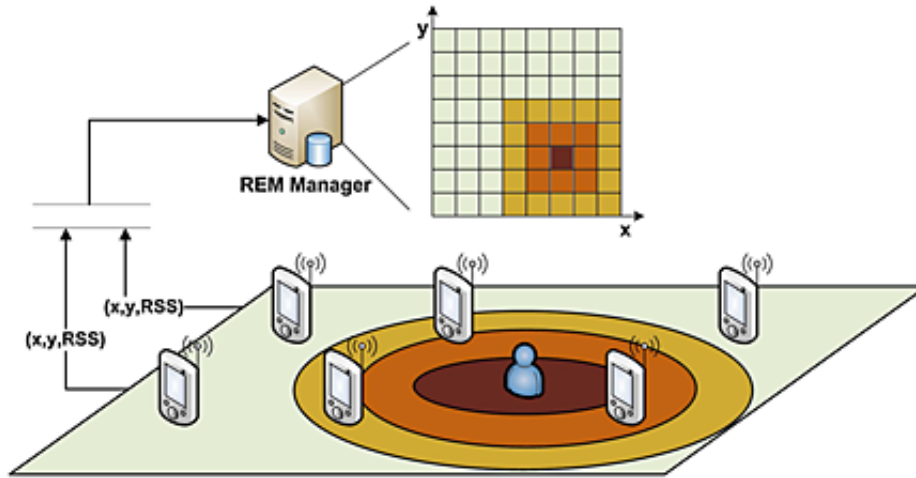


Figure 1.2: RSS-based REM construction [9].

REMs were introduced as a CR supporting technology. CR as technology was successful in sensing the available and occupied spectrum employing spectrum sensing capabilities. However, its evolution for practical scenarios was limited to only monitoring the spectrum for opportunistic access [1]. As an enabler, REM could not only overcome the limitations of a CR-supporting device such as base stations (BSs) or mobile users but also sense and estimate the dynamic nature of the radio environment. This led REMs as a technology to evolve as the trend to accommodate more use cases since the development of current and future cellular and wireless technologies. REMs sense the RSS measurements from a dedicated WSN using distributed spectrum sensing to leverage the information contained in the environment [7]. Therefore,

distributed spectrum sensing is an important part of a CR. It benefits the REM by capturing RSS measurements without extra hardware, as it is supported as a default in most sensor nodes or sensing units. Through this work, we look to build a power map employing RSS-based localization and compare the benefits of REM maps in 5G use cases.

1.2 Main objective

The master thesis evaluates the performance of the two transmitter localization algorithms, namely Location estimation-based (LiVE) and Mean squared error (MSE) based methods in a distributed sensor network, where the dedicated sensors capture the power transmitted by the unknown transmitter or an emitter for performing the localization. The estimated transmitter location is then used to calculate the estimated power levels at the sensors for constructing a REM. The objectives of the thesis are listed as follows:

1. Implementation of the localization as mentioned above algorithms and comparing the performance using performance metrics as follows:
 - Localization error and accuracy of the localization in meters.
 - Computational time to evaluate the performance of the two localization methods.
 - Limitations in terms of the spatial distribution of the sensors from the transmitter, size of the area of interest.
2. Modeling of large-scale fading effects, i.e., shadowing for two cases:
 - Classical log-normal shadowing, i.e., independent links between the transmitter and the sensors or links between the sensors.
 - Correlated shadowing, as in reality, the links at the transmitter and the sensors and vice versa are correlated in space after a certain decorrelation distance.
3. REM construction based on the evaluation of the localization methods to generate power maps to characterize the environment.

1.3 Organization of the thesis

The organization of the thesis is as follows, the first chapter provides a brief introduction to the topic. The second chapter contains the fundamentals and main concepts to provide a thorough explanation of the thesis objective and problem statement. Chapter three discusses state-of-the-art REM use cases and application-based REM generation. Chapters four and five discuss the thesis's implementation aspect, including localization algorithms, channel model, system model, extension to system model (specifically extension to transmit signal processing), and preliminary results. Chapter six concludes the discussion and scope for future work.

2 Basics

Chapter two offers a basic introduction to the fundamental concepts of the thesis. We begin with a detailed overview of distributed spectrum sensing and its significance in the sensor network for sensing RSS measurements. The importance and relevance of sensing power measurements are discussed in the context of REM map construction. We discuss why REMs are relevant for monitoring the radio environment and their applications. Further, we narrow down the construction of REM maps employing only the RSS measurements received on the sensing nodes or sensing units. These RSS measurements are essential for the localization of a transmitter active in a frequency band, providing us with additional information about the radio environment and its importance in REM construction. We conclude this chapter with state-of-the-art methods for RSS-based transmitter localization.

2.1 Sensor network model

Sensor nodes are a technology that enables the use of applications without management. Many sensors deployed for a specific use case, for example, localization, monitoring temperature and humidity, low power tracking devices, locating jammers or interferes, etc. are becoming important for IoT-based applications. In the considered scenario these sensors act as sensing units in a given pre-defined area, and we use their power measurements for localization. These devices could be low cost, low power consuming, and only capable of power measurement for high-end spectrum analyzers capturing power and spectrum in use. For ease of implementation, we consider sensors as sensing units.

The following sections discuss using sensor nodes in distributed spectrum sensing and RSS-based localization applications.

2.2 Distributed spectrum sensing

Depending on the system design, a spectrum sensor does not necessarily decodes transmissions into messages, as it is primarily not a coherent receiver. The aim of the spectrum sensor is to characterize the radio frequency (RF) environment. We require knowledge of time, frequency, and geography (terrain information) to capture power measurements at the sensing units.

Spectrum in use characterizes the following information [10]:

- Time - temporal characteristics of spectrum use.
- Frequency - occupied bandwidth and frequencies.

- Location - geolocation where the electromagnetic waves propagate.
- Angle - beam angle or phase information of the transmitted beam.

Spectrum sensing implies the use of just one sensor to sense the data and characterize the spectrum in use [10]. This leads to several problems, such as unreliable sensing data, noisy measurements, and interference which leads to erroneous spectrum characterization. Similarly, centralized spectrum sensing systems have only one sensor, while distributed spectrum sensing systems have multiple sensors. Systems with multiple sensors present interesting opportunities for spectrum sensing, increasing the implementation challenges [11]. Distributed spectrum sensing can enable the collection of data that varies by location. This compensates for the primary disadvantage of having only a single sensor: inaccurate localization or impairments. However, disadvantages of using distributed spectrum sensing are: intra-sensor communication and data fusion.

With distributed sensing, the data collected at each sensor varies geographically and may vary by time depending on the accuracy of the sensor's clocks [12]. In figure 2.1, data of sensors A and B is affected only by the surrounding geography, since the two sensors and the transmitter. In figure 2.1 (a), this is of little consequence, the two sensors are roughly equally placed away from the transmitter with no obstacles. However, in figure 2.1 (b), an obstacle between the transmitter and sensor B causes shadowing implications, attenuating the signal and the data from that sensor.

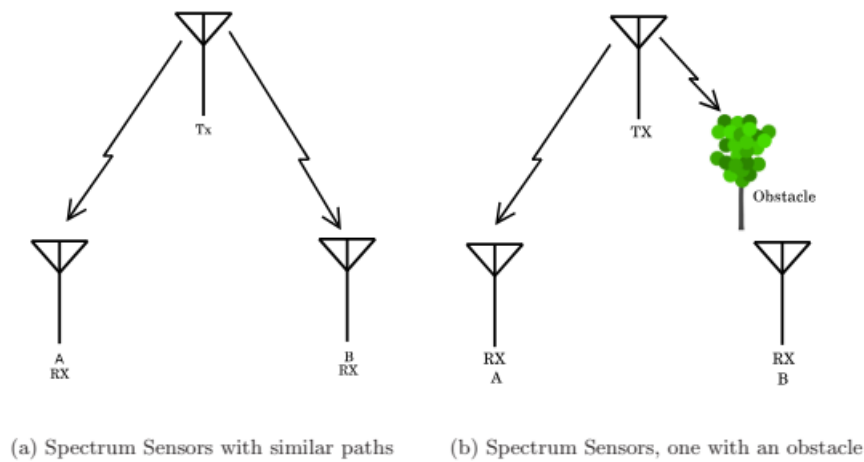


Figure 2.1: Distributed spectrum sensing for two sensors affected by environment [10].

Distributed spectrum sensing enables spectrum to be sensed from multiple channels or links. Sensing from multiple channels allows the spectrum sensors to work together to circumvent channel impairments, such as shadowing that are local to one sensor or a network. We use an arrangement of dedicated sensing units to capture the power measurements affected by free space path loss and shadowing impairments. However, improving system performance is not always as straightforward as adding sensors to the network. More sensors do not necessarily mean that spectrum sensing capabilities will increase. Additional sensors experience correlated shadowing or fading, resulting in poor performance if the underlying properties such as correlated

shadowing and fading impairments, are not accounted for [13]. For improved spectrum sensing performance, the channel that each sensor senses must be independent of all the other channels. Sensors that are separated by a greater distance are more likely to observe independent links. Fewer sensors that are more spread out in a geographical space are more effective than a larger number of sensors condensed into a small space because those sensors account for similar fading. The surroundings of the deployed sensors play a vital role in sensing measurements. For example, the two sensors in figure 2.2 (a) are inhibited by an object and show correlated shadowing, while the sensors in figure 2.2 (b) have independent channel paths conforming to i.i.d. log-normal shadowing behavior.

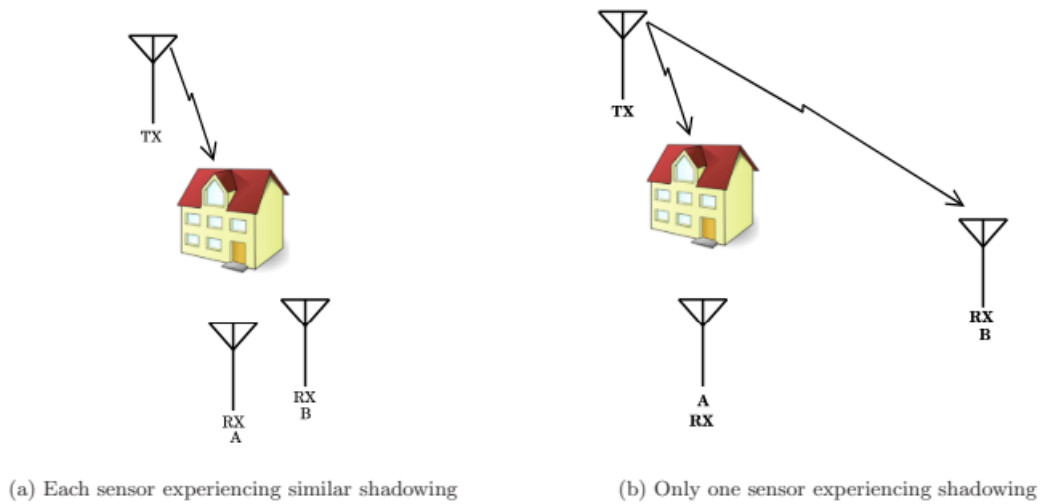


Figure 2.2: Sensors affected by shadowing objects [10].

The impact of correlated shadowing compared to i.i.d shadowing will be discussed in detail for our localization performance task later in Chapter five.

Additionally, we make use of the measurements from our sensor network at the data fusion center, which collects and fuses all the data from the dedicated sensor network for further processing in an offline manner. This data fusion center uses the collected data and performs vital tasks such as localization (the goal of the thesis) and spectrum analysis.

2.2.1 Comparison of centralized and distributed spectrum sensing

We conclude from the comparison table 2.1 that using a dedicated sensor network gives us robust measurements for our localization task compared to classical spectrum sensing. Next, we briefly discuss the RF sensing chain of these sensors and sensor design.

2.2.2 Sensing RF chain and generic sensor design

The system should be able to make wideband, spatially varying, and time-varying measurements. The system should also have the mentioned attributes of practicality and ease of use. In our case, we employ the use of a wideband scenario for these sensors to capture the data, process the data,

Topic	Centralized	Distributed
Fading and Shad- owing	Cannot detect or correct	Can detect or cor- rect
Implementation Complexity	Low	High
Sensitivity Requirement	High	Low
Scalability	None	Can scale

Table 2.1: Comparison of Centralized and Distributed Spectrum Sensing [10].

and send the data to the data fusion for post-processing. Examples of devices that could be used are spectrum analyzers or standard USRP devices.

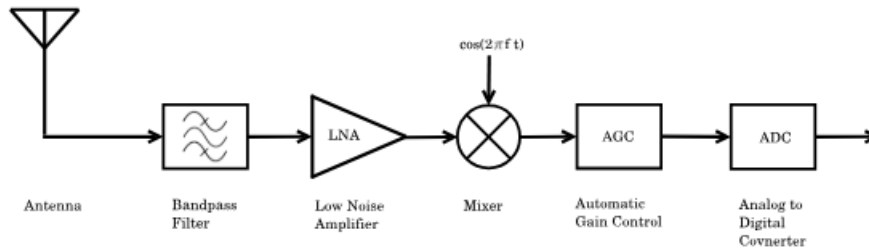


Figure 2.3: A generic RF signal processing chain [14].

In the given generic RF chain, there are several components each with its own importance for the received signal power at the receiver side. The antenna receives the signal, it passes it to the bandpass filter to remove the unwanted frequencies. The low noise amplifier (LNA) amplifies the signal above a noise floor. The mixer downsamples the signal to the baseband and passes it to the ADC to convert the baseband to digital symbols.

A selection of a suitable application-based sensor is based on factors such as [10]:

1. Low cost with power capturing ability.
2. Each sensor must have wideband sensing capabilities, from 50 MHz to 2.2 GHz.
3. The sensor should be independent, that is the sensor should not need outside assistance after the initial setup or before the end of the scan. Each sensor should be able to function independently for an extended period of time.
4. The system should provide a mechanism for the sensors to share data via a computer network or offline.

2.2.3 Sensing modes

The sensing modes define the nature of the transmissions (waveform, training/sensing sequences), spectrum, and other resources utilized for sensing. Sensing occurs with dedicated signals for sensing or exploiting the available message signals. There are five sensing modes that can be chosen based on the information and application requirements.

- Instantaneous/On-demand mode makes an on-demand, singular, or short burst of measurements.
- Opportunistic mode exploits the received and/or transmitted communication signals for sensing [10].
- Passive mode exploits sensing and/or communication signals transmitted by other devices (to each other) for sensing.
- Periodic mode takes measurements with a certain frequency/rate, whether from the available communications signals or dedicated sensing transmissions.
- Threshold-based mode only initiates a complete sensing process or forwards the observed data if the initial measurement passes a certain value

Here, the threshold-based mode is optionally used alongside another sensing mode as a regulator for forwarding sensed data or frequency of sensing. Additionally, periodic and instantaneous sensing modes may require feedback [36], depending on which node is transmitting the sensing transmission. Depending on the application scenario, certain parameters can be associated with each sensing mode. These parameters could be sensing rate f , sensing period T , update threshold T_h , sensing waveform, and its parameters, such as bandwidth and power. The REM manager or sensing application chooses the parameters based on the application performance metrics, such as accuracy.

In chapters four and five, we discuss in detail how we may use these power measurements for REM construction and RSS-based transmitter localization.

2.3 REM construction

A geolocation database is a collection of information about a radio environment. This database contains the following information: terrain or geography, transmitter and receiver positions, and channel conditions. This set of information helps a network operator to construct a real-time overview of the spectrum and frequencies in use in a given area. A CR technology-supporting device senses the captured data using distributed spectrum sensing in a given location and processes these measurements using data fusion in a centralized server. A network operator benefits from these captured measurements in a way such that the measurements are sent to a REM manager which processes the data from these sensing units and helps the operator characterize the radio environment. REM helps us in identifying more parameters than a geolocation database such as transmitter and receiver positions, spectrum in use, propagation channel properties, and duration of spectrum use for primary users as in CR but also helps in identifying interference, power distribution, channel gain properties, communication signal

waveforms and modulation schemes used in a radio environment. Additionally, a dense wireless sensor network heavily strains the bandwidth and achievable data rate for users assigned in a spectrum. In such conditions, an intelligent and dynamic method such as REM is essential and vital for efficient resource utilization.

As described above, these maps make use of the channel parameters and construct power maps, interference maps, and channel gain maps. An important use case for this thesis is constructing power maps to identify active transmitters in a given time and frequency of operation. In a nutshell, REMs are an extended and intelligent version of a geolocation database. Figure 2.4 gives a simple block diagram for REM architecture.

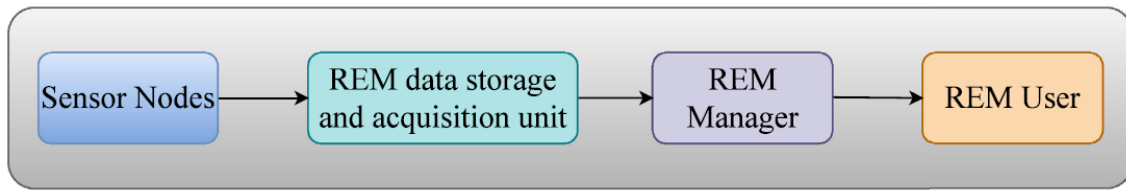


Figure 2.4: REM block diagram [15].

An important aspect of building power maps is active transmitter localization. Localizing the unknown transmitter provides us with vital information about the actual transmitter transmitting in a particular frequency and gives us knowledge about interferers and jammers, which usually inhibit the spectrum and are harmful to the network. However, the focus lies only on single transmitter localization and building power maps. Interference map construction is out of the scope of the master thesis.

In the following sections, we briefly mention the functional blocks of a REM system and their fundamental functionalities in processing the captured data from the sensing units.

2.3.1 Essential functional blocks of a REM system

As shown in the following figure 2.5, we take a closer look at each of the functions of a REM system in the following subsections. We briefly start with the geolocation database, and then discuss the generic REM data model for CR-supporting devices and conclude the section with REM construction methodologies.

2.3.1.1 Gelocation database

Geolocation database keeps the information about primary users (PU)s (possibly CR-capable devices), other transmitters, and receivers. However, a geolocation database contains static information about the environment whereas a REM keeps track of dynamic changes in the environment. Therefore, REM is recognized as a cognitive or intelligent entity surpassing geolocation databases [9]. These databases provide three basic REM enabling services [7]:

1. Data repository: Database storage where permanent service, user information, and regulatory information for country-based operators are stored.

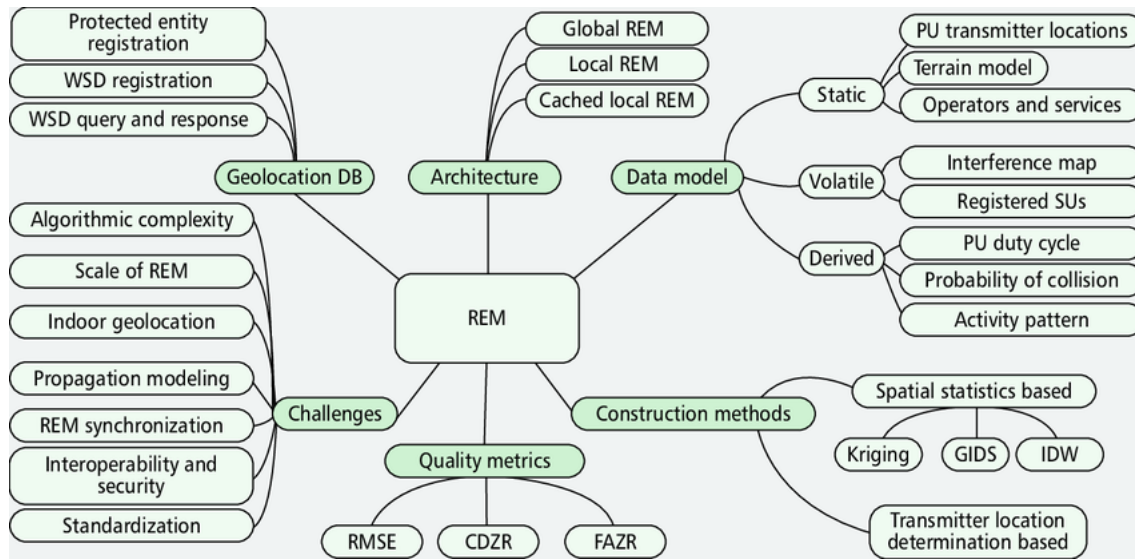


Figure 2.5: Functional blocks of a REM system [7].

2. Registration: Consists of registration of protected entities such as TV White Spaces and the CRs devices.
3. Query and response: Keeps the list of available channel lists for both the CR and the public interface.

2.3.1.2 REM data model

REM addresses the dynamic and multi-domain range information about the radio environment, which is harder to acquire for individual CR supporting devices. The multi-domain information is categorized as [7]:

1. Radio resource elements: information about the transmitters and receivers in the area.
2. Environment: information about the channel conditions, interference map, power maps, and geography.
3. Radio scene: Provides information about the policies, and regulations (country-specific) and operators in an area.

Different types of REM maps can be constructed based on the type of data acquired. The data can be classified as static, volatile, and derived. Different spectrum maps are associated with the acquired datatype as shown in figure 2.5.

2.3.1.3 REM data processing

After data acquisition, the next step is what to do with that data. Data processing is a very important aspect of REM construction as it helps us to identify various parameters of REM through rigorous analysis. Data processing is taken care of by the REM manager, then the overall design of a REM model involves three steps [16]:

1. Data gathering and representation: The sensors or sensing units compile a report consisting of received power levels, frequency in use, and positions of devices and send them to the REM manager [17].
2. Data fusion: REM manager fuses the gathered information for possible decisions on spatial statistics for the radio environment modeling, transmitter and PU activity.
3. Data query: CRs consult the REM manager about the possible operation parameters in a region of interest. For example, REM holds the information about current and historical data of a region and queries it whenever required in case of major policy or radio environmental changes [18], [16].

A flow diagram visualizing the design of a REM internal structure is described below in figure 2.6.

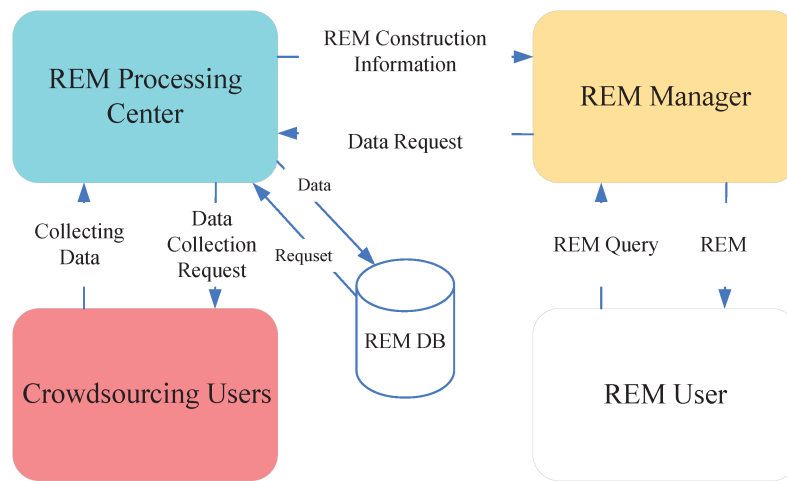


Figure 2.6: REM data and design flow diagram [19].

2.3.1.4 REM construction techniques

Once the REM manager acquires the measurements (received power) and processes them in the data fusion center, we focus on the last step, REM construction. REM construction depends on how fast or frequently the environment changes in a periodic manner (frequency of data collection) or remains unchanged for rural scenarios with less frequent changing statistics compared to urban scenarios. We focus our REM construction in a given area where each point is considered as a pixel in discrete space.

In the literature, REM construction methods are classified into two types: spatial statistical methods and transmitter location determination-based methods, also called direct and indirect methods, respectively [6]. These methods are shown in figure 2.5.

Spatial Statistics Based Methods: As the name suggests, this method describes the spatial statistics of the given region of interest. These methods take into account the statistics and measurements at certain locations of interest and use interpolation techniques to estimate for example, power measurements at locations where measurements are unavailable. Classical

algorithms that employ the use of this method are kriging, inverse weighted distance, and nearest neighbour methods. Authors in [9] make use of these algorithms to construct REM methods in an area spanning 1000m x 1000m. Important aspects of these methods are the frequency of measurements and the density of sensors deployed.

Transmitter Location Based Methods: Spatial statistics based methods require a precise and high volume of measurements to provide an accurate estimation of transmitter positions. The spatial statistics method does not require the knowledge of the transmitters or power source for estimating the signal strength measurements at places where measurements are not available to construct a REM. However, transmitter location-based methods make use of the information of source power to make the problem of reconstructing REM easier and more readily available. This method involves knowledge of the propagation channel model and source power to estimate the transmitter or even jammers in a region of interest. Therefore, the work and goal focus completely on transmitter location-based methods using RSS measurements, providing us with multiple degrees of freedom such as multiple transmitters, source powers, and propagation channel characterization [20]. We compare the performance of two indirect methods, namely: LInE and MSE methods in Chapter five.

2.4 Transmitter localization methods

Localization is an important aspect of CR networks and state-of-the-art cellular technologies such as 5G and 6G in indoor scenarios. Localization finds its application in IoT-based use cases, ultra low latency communication (URLLC) scenarios such as vehicle to anything V2X or cellular to anything CV2X, military based applications for security and defense, etc. In principle, any signal propagating in a wireless environment intrinsically conveys position-dependent information that can be exploited for localization from underlying control signal messages loaded inside the symbols. Such position-dependent information can be extracted from measurements campaigns such as RSS, time-of-arrival (ToA), angle-of-arrival (AoA), phase, or combinations of them, depending on the radio technology [21]. A dense network of wireless sensors computes signal measurements with respect to one or multiple reference transmitters and then infers the position by means of a localization algorithm.

In the realm of cellular networks, the sources used as reference transmitters encompass navigation satellites, cellular (BSs), and, in the context of cooperative localization, other mobile users. Historically, satellite navigation has held the forefront of localization technology due to its global coverage and impressive use of global positioning systems. To complement this, cellular-based localization serves as an additional solution for situations where satellite signals are obstructed, commonly occurring in urban and indoor settings. Positioning techniques fall into two primary categories:

1. Mobile-based methods involve the (mobile) device itself calculating its own location by utilizing signal measurements from both terrestrial and satellite transmitters. Network-provided assistance data can also be harnessed to conduct signal measurements and deduce the device's position.

2. Network-based approaches revolve around the network's location server deducing the mobile device's position through signal measurements. We employ the use of a network-based approach for our case of transmitter localization using a dedicated sensor network.

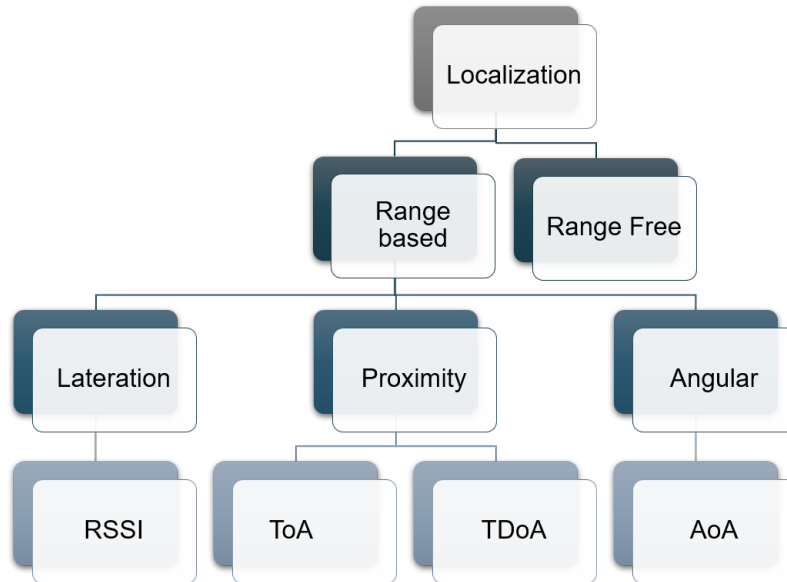


Figure 2.7: Classification Localization techniques [22].

As shown in the flow diagram 2.7, the localization is divided into two main categories with our focus on the range dependent methodologies. The range-dependent algorithms are divided into the following subcategories [21]:

- Trilateration and Multilateration: The position estimate is obtained by intersecting geometric forms, e.g. circles or hyperbolas, created by distance or angle measurements between the terminal and the reference transmitters or receivers. In our case, we focus entirely on RSSI-based localization [21].
- Proximity: The unknown transmitter position is estimated using ToA or TDoA methods [23].
- Fingerprinting: The algorithm is based on finding the best match for a certain signal measurement, such as RSS, time delay, or channel delay spread, from a database of fingerprints. Each fingerprint is associated with a specific location.
- Angular: Mainly focusing on antenna beam pattern based or AoA-based method [22].

In the succeeding subsections, we briefly describe the ToA, TDoA, AoA, and RSS-based methods.

2.4.1 Time of arrival

ToA technique [24] is mainly based upon accurate measurements of times of arrival transmitted and received signals between two sensor nodes. Distance information between two sensor nodes is provided by ToA, which estimates the flight time of a signal from one node to another. The

distance between the i -th access point (AP) and the transmitter using the time of arrival or delay tags can be estimated using the equation:

$$d_i = (t_i - t_0)c \quad (2.1)$$

where the time instants t_0 and t_i is the time of signal transmission and time of signal reception respectively. The following figure 2.8 provides us with a clear description of ToA method.

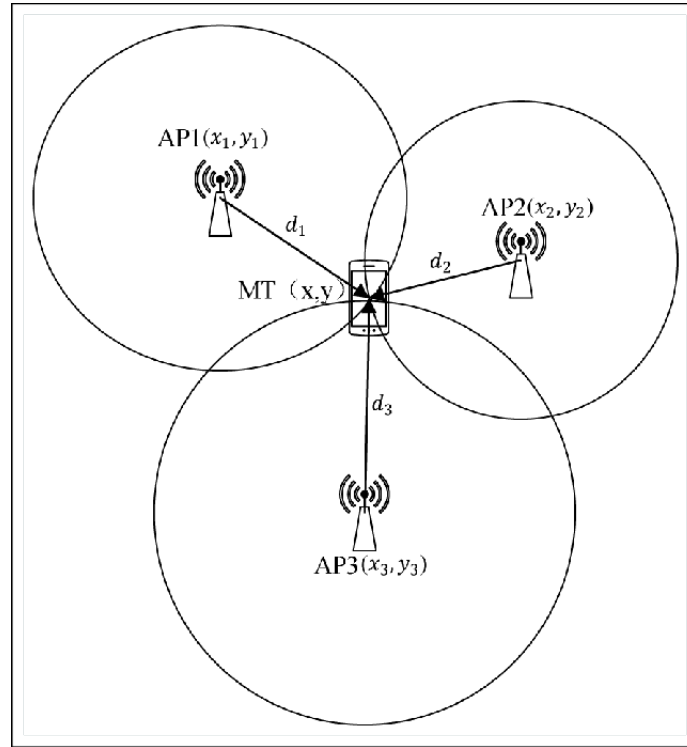


Figure 2.8: Trilateration using timne of arrival [25].

2.4.2 Time difference of arrival

It is similar to ToA, but sometimes the time is unknown, so the difference between travel time is used. The intersection of the hyperbole of the differential ToA of the signal estimates the transmitter location.

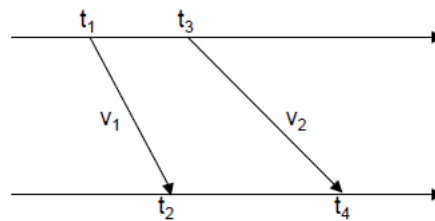


Figure 2.9: TDoA time based in linear curve.

A signal transmitted at time t_1 is received at time t_2 followed by a delay $t_{delay} = t_3 - t_1$ and the signal transmitted at time t_3 is received at time t_4 with velocities v_1 and v_2 respectively. The distance estimated by the receiver is given by the equation:

$$d = (v_1 - v_2)(t_4 - t_2 - t_{delay}) \quad (2.2)$$

Measurements of TDoA can be calculated using measurements of ToA of two sensor nodes as in Figure 2.10. Synchronization between transmitter and receivers is required in this approach.

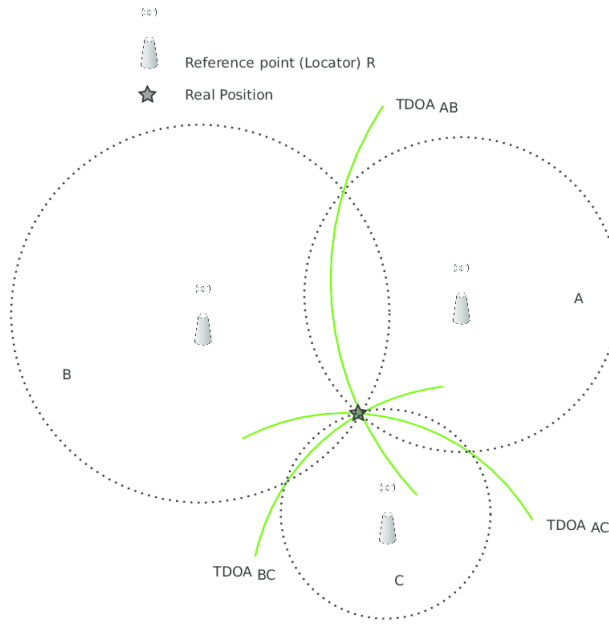


Figure 2.10: TDoA using two or more ToA measurements [23].

2.4.3 Angle of arrival

AoA [26] mostly rely on directional antennas or multiple antenna configurations to estimate the angles or phase of the received signal from the sensor nodes. The angle that exists between some referred direction to the propagated direction is known as the angle of arrival.

High accuracy in localization based upon measurement is possible in the AoA approach, for which we require antenna arrays with certain spacing for spatial diversity having higher complexities to measure the required direction. For specific WSNs, getting that much spacing between arrays may not be feasible considering the size of sensor nodes.

2.4.4 RSSI-based localization

RSSI-based localization in WSN has found its way into many literature research [6, 7, 16, 27, 28] because the knowledge of the source of power with appropriate propagation channel makes

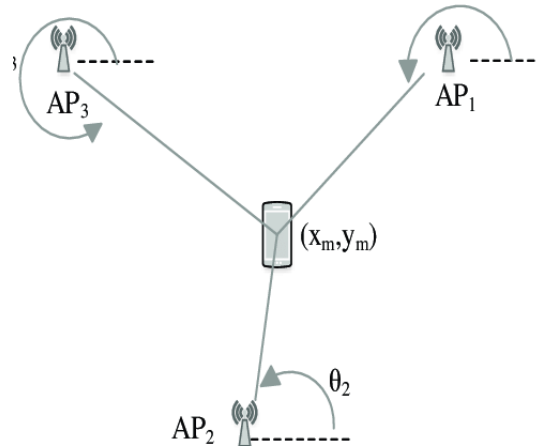


Figure 2.11: AoA scenario [26].

it easier to localize the unknown transmitter for CR and spectrum sensing use cases. However, the RSS-based localization is limited in scope for CR supporting technologies and military use cases, indoor positioning for 5G and beyond, NB-IoT and unmanned aerial vehicles (UAV) based transmitters, etc. The number of applications keeps increasing because of its practicality.

As per the above depiction of the three methods (i.e. ToA, TDoA, and AoA), it is obvious that RSS-based techniques possess many more advantages, including no/ less requirement of extra hardware with less complexity and cost. RSSI measurement is supported by actual sensors (Imote2, TelosB, MicaZ, WSN430, etc.) and 802.11 and ZigBee standards [3].

The working principle states that the transmitter is close to the sensor with the highest received power levels. Therefore, the received power levels are described as a function of the distance between the sensors, operating frequency, path loss, or attenuation constant, which is environment dependent, unknown transmitter power P_t , and antenna gain G respectively. The function formulation is described as follows:

$$P_r = f(d, \lambda, \alpha, P_t, G) \quad (2.3)$$

Possible problems with RSSI-based localization methods include:

- Fading: Large-scale and small-scale fading affects power levels effectively.
- Shadowing: Depending on the environment urban or rural shadowing (i.i.d. or correlated as in real-world scenarios) impacts power measurements.
- Path loss: Path loss exponent varies heavily as the scenario changes dynamically, and the use of the appropriate model for channel propagation is complex to find and implement, as shown in figure 2.12 [29]

RSS has some challenges in terms of accuracy and stability. Hence, to improve accuracy some of the features need to be taken into consideration:

- RSS needs to be measured over several operating frequencies.

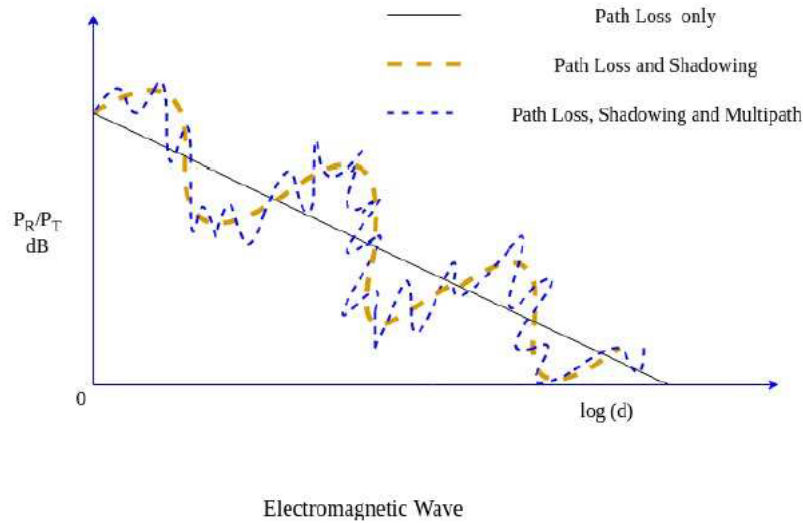


Figure 2.12: Path loss variation vs distance of propagation [29].

- For smooth variations, the average of RSS values needs to be considered [3].
- Sensor radio needs precise calibration.
- Quality of the sensor antenna must be good.
- Interference must be minimized.

RSS-Based transmitter localization in WSNs with DSS's applications can be summarized as stated below :

- **Optimizing spectrum sensing nodes:** In a WSN focused on DSS, nodes are deployed to monitor and detect wireless signals in a given area. Knowledge of locations of these nodes through RSS-based localization helps in optimizing their deployment for effective spectrum monitoring.
- **Interference mitigation:** In RSS-based localization, WSNs can identify and mitigate interference sources in real-time, enhancing the reliability of spectrum sensing.
- **Dynamic spectrum allocation:** RSS-based localization can assist in dynamically allocating spectrum resources in a network. For example, it can help identify available frequency bands and optimize frequency assignments for different use cases.
- **Security:** In WSNs with distributed spectrum sensing, knowing the location of potential threat sources (e.g., unauthorized transmitters) through RSS-based localization is crucial for security monitoring and threat detection.
- **Healthcare:** In healthcare facilities, indoor localization is used to track the location of medical equipment, staff, and patients, improving asset management and patient care.
- **Indoor navigation:** RSS-based indoor localization is crucial for indoor navigation applications in places like shopping malls, airports, hospitals, and large office buildings. It helps users find their way and locate points of interest.

- Automated warehouse; Another important application is locating automated robots in warehouses or even in industrial applications to monitor the robots movement and environment of operation.

An intensive and detailed view of RSS-based localization is carried out in Chapter 4 where we mention path loss modeling and indirect REM construction methods and compare their performances.

2.5 State of the art and related work

In this section, we introduce some sota and related work in technologies from the literature review, comprehensively studying the localization methods used for REM construction. We divide the sota into two parts: RSS-based localization and REM construction. Firstly, we start by making a case for the various localization algorithms used in the literature and then introducing the algorithms we base our comparative study on. Secondly, we deep dive into the REM construction methods and make a case for REM in 5G.

2.5.1 RSS-based localization

The traditional localization algorithms using ToA, AoA, and TDoA require comparatively more system parameters, such as a high volume of measurements, antenna arrays, time synchronicity, accuracy of data, etc., that we overcome by the use of low-cost and only power-consuming RSS measurements. The authors in [9] introduce a location estimation based localization algorithm for REM construction. The method is a manipulated version of the path loss model to estimate the unknown transmitter location in a given sensor network. However, they do not mention the algorithm's limitations in terms of accuracy amongst other localization methods, such as inverse weighted distance (IWD) and kriging interpolation. The authors in [6] formulate a least squares-based linear regression problem for localization called a MSE-based approach. They claim the algorithm has an accuracy of less than 1 m for a scenario that is (35m, 35m). However, they fail to address the algorithm's performance in a general scenario, which we evaluate later in Chapter five.

In [30], the authors implement an indoor localization scenario with a mobile transmitter and many obstacles. They introduce three filtering methods to increase the accuracy of the IPS detection, namely, feedback filter (FF), modified unscented Kalman filter (KF1), and modified Kalman filter (KF2), and evaluate their performance based on MSE metric. They conclude that KF2 performs the best for indoor localization with an MSE value of 1.09. The limitation of the research was the use of at least three nodes for capturing the measurements. The research employs a Kalman filter to mitigate the noisy environment, as RSS measurements inherently incur noisy measurements because of obstacles in the enclosed space.

In recent studies [28], particle simulation is seen as a powerful tool for accurately localizing an active transmitter or multiple transmitters. The authors use a correlated shadowing model to conform to real-world scenarios. Similar to Kalman filters, which are the primary algorithm

for tracking tasks, particle simulation-based tracking for mobile scenarios is a very promising algorithm.

In addition to the classical range-based algorithms, recent work focuses more on machine learning (ML) and deep learning (DL) aspects for the localization of single and multiple transmitters. In [27], authors compare compressed sensing and REM localization-based approach for localization incorporating correlated shadowing. In compressed sensing (CS), the knowledge of transmitter power is essentially not required as in other methods. Based on the simulation results, the authors infer that the CS is better in localization but performs less than kriging regarding power reconstruction. This research reveals several important aspects while choosing the algorithm based on the application.

In [31], the authors introduce the use of Gaussian mixture models (GMM) to implement probabilistic based models. In reality, wireless channels (Rayleigh fading models) are not Gaussian, but using a mixture of gaussian random variables gives a better representation of the model.

Several deep learning-based models are presented in the literature [27, 32, 33]. When dealing with an extensive RSS database or a complex environment, a conventional fingerprint ML algorithm's computational time increases, increasing the run time. The RSS-based localization using the deep learning-based feature extraction method first involves pre-processing RSS values before feeding them into the feature extraction method [22]. The pre-processing involves normalization, calibration, augmenting, and dimensionality reduction using statistical methods or traditional ML algorithms such as support vector machines (SVM). A crucial aspect of DL-based learning is the need for accurate data and algorithm parameter definitions. Several research articles provide a detailed view of the parameters involved and the collection of datasets for more precise results [32], [33], [15]. However, the discussion of the thesis is limited to introducing such algorithms rather than a detailed overview.

Another fingerprinting-based localization scheme in [34] outperforms GMMs, SVM, and K nearest neighbours is random forest. This method provides higher accuracy in the testing phase.

In [35], emphasis on localization based on 5G technology where industrial, commercial, and academic solutions were summarized. Although [35] moderately covered practical limitations, theoretical limits were not addressed [36].

Localization based on classical methods and technology (4G) was considered for static transmitters, receivers, and mobile transmitters. However, the deployment of 5G and its use cases, such as ultra low latency and reliability (URLLC), low latency, mobility, and high data rate, bring more complex scenarios into play, for example, vehicle to anything communication, and UAV-based communications which require seamless and reliable transmission. The challenges incurred with these use cases are reliable communication, latency, and mobility (Doppler effect), making compensation harder for WSN-based static scenarios. Therefore, we require localization schemes that can accommodate 5G functionalities.

An exemplary use case scenario of UAV, which employs various 5G functionalities, is discussed to understand the challenges incurred in localization aspects. The state-of-the-art localization schemes considered in the literature use RSS-based measurements. However, in 5G, the use of a large number of antennas (Multiple Input Multiple Output (MIMO)), Doppler, reliability

of measurements, synchronization, beamforming for high-frequency communications, timing delays, etc. are of higher importance and cannot be fully exploited by RSS-based localization methods but need considerations for TDOA, AoA and ToA based schemes to provide better localization accuracy. In other words, state-of-the-art applications involving 5G use cases are susceptible to demanding and accurate measurements which are affected by the technology used for capturing the data such as ultra-wideband (UWB), synchronization of the deployed sensors for a moving object, beamforming for target localization or signal transmission, delays caused by non-line of sight (NLOS) or multipath propagation.

Some of the methods that encompass the use of 5G features for localization are bayesian filters amongst the most used methods for mobile robots localization, unscented Kalman filter, particle filter simulation, extended Kalman filter, and different variants, such as square root cubature Kalman filter (SRCKF) and diffusion extended Kalman filter (DEKF) [37].

Another application for 5G-based localization is robotic and sensor fusion or indoor robot-based sensing. In this application, we employ the use of 5G waveform (OFDM), which employs the use of either a positioning reference signal (PRS) in the downlink or a sounded reference signal (SRS) in the uplink. This application is heavily dependent on sensor synchronization and beamforming in near-field scenarios, which corresponds to higher accuracy in the localization of robots.

The localization aspect is more incorporated towards angular and time difference of arrival-based measurements instead of RSS measurements as a straightforward requirement of these applications leans towards beamforming and phase errors.

Chapter 5 describes a thorough discussion of OFDM waveform-based localization. In Chapter 3, we look into more detail about UAV applications and how localization is a boon for REM construction and REM update rate for mapping changes in the dynamic environment of UAV-based applications.

2.5.2 REM construction

In this section, we discuss and name a few spectrum cartography or power map construction techniques based on transmitter localization and power construction algorithms, as mentioned in the previous section.

In [27], authors introduced a REM localization-based method and CS based method for multiple transmitter cases. The salient feature of REML and CS approach is the assumption made by REML over the multiple transmitters transmitting with the same power, and no such assumption is taken into account for CS approach as it estimates the location and the powers. The authors point to an interesting conclusion regarding the performance of REML and CS in terms of localization and REM construction. REML performs better during REM construction but cannot separate individual transmitters as the power regions tends to merge because of the comparable power levels. However, the performance of CS approach remains unchanged for both localization and power-level construction. In conclusion, REML performs better than CS in terms of power-level reconstruction.

In [15], the authors provide a detailed survey of CS-based REM construction methodologies. In [7], introduced a REM construction based on the LlvE algorithm, one of the most prominent in the literature based on received power level reconstruction using interpolation methods.

In dense urban areas, the shadowing objects attenuate wireless signals. To address this issue, a use case with the deployment of UAVs as relays to forward the information between BS and user equipment (UE). However, the position of UAV which acts as a virtual BS affects the channel gains between BS and UE. In order to capture the signal strength and propagation conditions, the authors in [38] have exploited the finely structured radio map. Using machine learning is crucial to reconstruct the radio map accurately, as the impact of shadowing objects limits the volume of captured signal strength measurements. For spatial spectrum sharing in the CR network, a probabilistic interference-constrained method with a REM for spectrum sharing has been proposed in [39]. The authors employ an ordinary kriging method to construct REM and estimate the power level differences at places without available measurements. Subsequently, the proposed method in [39] has been compared to the existing path-loss model, perfect estimation, and kriging-based, without error prediction. The proposed method outperforms the abovementioned methods, even with fewer data measurements regarding spectrum-sharing opportunities.

In [40], Shepard's interpolation technique has been considered to construct a spectrum map by fusing the information shared by SUs, considering their mutual distances and spatial orientation with respect to each other. The spectrum map informs us of the spectrum occupancy in a region of interest. The authors propose using a spectrum map to estimate radio and network performance metrics like channel capacity, network throughput, and spectral efficiency. In order to localize the primary transmitter occupying the spectrum, a vector clustering technique that uses tree-structured vector quantization has been developed to determine the optimal locations of the SUs.

In [41], the authors refer to channel gain map reconstruction using Kriged Kalman filters. It considers using spatio-temporal properties of correlated shadowing and builds channel gain maps. We only refer to this research as a possible use of REM, but our focus entirely lies on spectrum or power map construction.

A cooperative spectrum sensing-based REM for localizing multiple transmitter localization was implemented in [9], which was used as a reference, and an extended version in [27]. Both these articles refer to REM maps for sensing active transmitters in an area and use power map construction as a means to localize multiple transmitters using interpolation techniques such as kriging method for accurate results.

With the advent of the successful applications of machine learning techniques in various research areas, it is vital and promising to devise learning-based radio map construction methods, given the large number of measurements needed for interpolation techniques. Instead of leveraging the signal propagation function in the model-based approach (estimating certain signal processing-based parameters and modeling the REM as a regression problem) or the interpolation function in the above model-free approach, the machine learning-based approach intends to find the direct mapping from an input position to its output PSD measurement, while treating the explicit functional relationship as a black-box, for example, modeling it by a DNN [42]. Employing a supervised learning-based method, available coordinate-PSD data pairs can train a DNN for regression analysis, approximating the coordinates to true PSD mapping. After training, the

trained DNN can quickly respond to any input location coordinate with its corresponding PSD level [15]. However, because of the vast area, a radio map must cover, a lack of training samples (measurements) may impact the estimation accuracy of fully supervised learning approaches. A promising method to tackle the problem is a semi-supervised deep reinforcement learning approach [42]. Specifically, one can randomly sample a location coordinate and apply the DNN to approximate the coordinate-PSD mapping. Then, a learning agent can accept or reject the coordinate-PSD pair (a data sample) based on the available sensor measurements. An accepted sample is used to train the DNN in a supervised manner, for example, using the mini-batch stochastic gradient descent method [42]. As a result, the deep reinforcement learning approach can autonomously generate data samples for training the DNN and thus continuously improve the estimation accuracy as the iterations proceed.

The use of crowdsourcing for spectrum sensing shows great promise for regularly gathering spectrum measurements across a wide geographic area. However, this approach is susceptible to false spectrum measurements, which can significantly impact the accuracy of the data collected. Constructing a reliable REM remains an ongoing research challenge, but a new secure crowdsourcing method called SecREM aims to address this issue.

SecREM is a novel technique that can construct a REM even in the presence of falsely acquired RSS measurements. This approach relies on a small number of trusted spectrum measurements to assess the accuracy of measurements from mobile users. Over time, these measurements are gradually incorporated to create a highly accurate REM. Extensive simulation based on a real spectrum measurement dataset confirms the effectiveness and efficiency of SecREM.

One of the challenges of crowdsourcing-based REM construction is the potential for false measurements, particularly from untrustworthy mobile users. These false measurements are a result of faulty sensors or malicious intent. Standard REM construction techniques like ordinary kriging can be significantly distorted by a few corrupt measurements, leading to missed spectrum opportunities or interference with PUs. While cooperative sensing aims to determine whether a PU at a known location is transmitting, secure REM construction deals with the challenge of possibly forged local spectrum measurements when the locations and transmission activities of PUs are known.

The challenges incurred by unsecured REM through classical methods remain inapplicable. Inspired by the self-labeled techniques [43] proposed for semi-supervised learning, SecREM constructs highly accurate REMs from a few trusted measurements. SecREM-based REM compares predicted RSSs and reported RSSs. In every subsequent iteration, several remaining trustworthy measurements are selected for the next update. This process is repeated until a certain terminal condition is satisfied; and at this point, the remaining untrustworthy or false measurements are discarded, and the final REM is produced [43].

Chapter 3 expands the REM construction use case for 5G scenarios and application-specific REM construction survey. We look closer at military use cases, URLLC, mmWave, and V2X applications.

2.6 Chapter Summary

The chapter introduces fundamental concepts of distributed spectrum sensing, sensor arrangement, radio environment, WSN localization, and sota REM construction research. We first state the mentioned in detail, and in the following chapters, we introduce REM construction in 5G use cases and compare the performance of RSS-based transmitter localization methods. Low latency, high data rate, and high-resolution angle and time-based measurements are just a few of the new capabilities that 5G NR will bring to the table that potentially revolutionize localization and the application of robotics. However, research towards localization via 5G is still in its early stages. The distinctive qualities of the 5G NR can make RF-based localization one of the most popular robotic tools for secure autonomous navigation. Future research will utilize such underutilized characteristics and overcome the primary technological challenges to guarantee that 5G is seamlessly integrated into the present localization system.

3 REM construction for 5G use cases and applications

As mentioned in Chapter 2, REM provides an abstraction of a radio environment with tremendous opportunities regarding network planning for operators and resource allocations. A REM contains rich information about the spectrum in use and propagation channel models of a wireless network in a geographic location, which can be leveraged to improve the performance of existing wireless services and enable new wireless applications. This chapter discusses some promising wireless applications that involve REM construction for 5G use cases, such as mmWave, URLLC, V2X, security-based applications, and mobile base stations technology (UAVs and VBS-based). We begin with the use case of mmWave-based REM construction and update rate.

3.1 REM for mmWave

5G and beyond work in a high frequency range called millimeter wave frequencies. Due to limited bandwidth and densely occupied networks, future and current communications scenarios operate under such frequency range, and these mm-wave mobile communications systems are likely to employ cognitive radio techniques [44]. A common feature in these techniques is a REM, which contains a database of radio channel properties as a function of mobile station (MS) locations and channel parameters, facilitating the determination of the radio environment for smart resource allocations. Rudimentary propagation information in an REM can be “bootstrapped” using path loss models or site-specific propagation prediction. Since future mm-wave systems will require high-directivity beamforming [45], it is expected that REMs for such systems will require refinement and updating using information collected “on the fly” while the system is in operation. In some cases – for example, in terrestrial military tactical networks – a static REM may not be available, and the REM may need to be constructed entirely over the required area.

An important aspect of working in mmWaves is the Doppler shift and beamforming for fast-changing channel characteristics. Henceforth, a specific application that has attracted much attention is the rapid development of high-speed railways (HSRs) [46]. HSRs impose strict requirements for wireless access, and there arise high requirements for mobile data traffic from the passengers on HSRs. Existing wireless access technology of the long-term evolution for railway (LTE-R) cannot meet these challenges due to its limited frequency bandwidth of up to 20 MHz. A potential solution is the mmWave which exploits the frequency bands ranging from 6 GHz to 60 GHz to provide multi-gigabit data rate broadband radio access [46]. Initial research in the mmWave system with MIMO technology suggests an achievable rate up to 7 Gbps throughput in HSR scenarios. However, this solution induces a high Doppler shift in carrier

frequencies due to the speed of trains. In [46] and previous research in mmWave-based Doppler shifts, the Doppler shift can be as high as 104 Hz in the case of 30 GHz carrier frequency and speed of the train 360 kilometers per hour. This shift results in time-varying channels and affects the orthogonality between subcarriers in OFDM systems [44]. OFDM-based signal generation for 4G (LTE-A) and 5G for such applications helps in tackling and mitigating the impact of high-frequency selectivity of the channels in fast-varying environments of speeding trains. This time variance and frequency selectivity of channels promotes use of OFDM-based waveform for building REM and estimating the propagation environment. Because of the speed at which HST move, the frequency change caused by the Doppler effect in HST is significant, for example, 1350 Hz. It also changes rapidly, like 4.9 kHz/s in systems, when a train passes by a base station [47]. This high Doppler shift can negatively impact the long-term evolution LTE-R system that relies on OFDM as it is susceptible to shifts.

In [47], authors propose a REM construction method to adapt to the application of high speed trains for such a high Doppler effect. By exploiting features of HSTs, i.e., regular and repetitive routes and schedules, a REM can be easily predicted and constructed using field tests based on a predictable Doppler shift curve. A maximum a posteriori estimator (MAP) method is proposed to estimate the Doppler shift accurately based on information collected from a REM. It uses the estimation from REM as prior knowledge and exploits OFDM's cyclic prefix (CP) structure to provide a MAP estimation on probable environment changes. The information provided by REM could be valuable in estimating Doppler shifts along the railway, including the shifts and corresponding speeds at different locations. Implying that the carrier frequency, measured speed, and location at a point, we can estimate the Doppler shift based on the stored REM data. However, it is essential to consider that measurement errors are unavoidable and can limit the accuracy of REM estimation in real-world scenarios. A MAP scheme [47] is suggested to enhance estimation performance. This estimator uses REM estimation as a knowledge base and leverages OFDM's cyclic prefix (CP) structure to achieve MAP estimation.

Various schemes have been proposed for Doppler shift estimation in high-mobility scenarios. For example, the state-space-based Kalman–Bucy filter estimator (KBFE) works well when the Doppler shift changes in the shape of a sine curve. However, it is difficult for KBFE to track the fast-changing rectangular curve since the Doppler shifts follow a rectangular curve in fast-changing scenarios [47]. Another method for Doppler estimation is the LOS path-separated estimator (LPSE) [48], which is designed for the practical HST channel. The errors in channel estimation heavily impact its performance, and thus, it performs significantly poorly in a low SNR regime. Most of the existing Doppler shift estimation methods for HSTs do not exploit features of HST communications such as the regular, repetitive, and predictable train route [47]. These parameters of a general train route and scheduling are exploited in building a REM and possibly in adapting the update rate of REM as the train moves along its scheduled route. The REM construction takes information about the train parameters such as location, route, speed, and carrier frequency from the signals of access points and estimates the REM using GPS positioning from satellites. REM should include various channel parameters at a given location along the HST rail, such as the number of paths, amplitude, phase, delay, and angle of arrival of every path. These parameters can be fine-tuned offline.

The following figure shows an example scenario for HST and Doppler estimation.

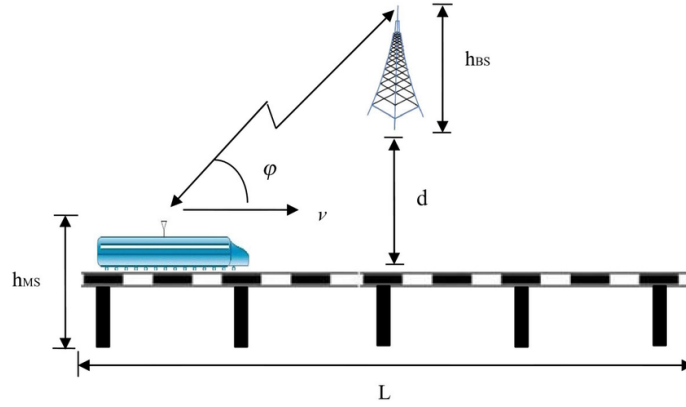


Figure 3.1: Doppler for high-speed trains in mmWave [46].

The following equation gives the Doppler frequency [47]:

$$f_d(x) = f_{carrier} \left(\frac{v \cos \phi(x)}{c} \right) \quad (3.1)$$

In equation 3.1, f_d is the Doppler frequency shift as a function of the train location, $f_{carrier}$ is the carrier frequency of the OFDM symbol, v , and c are the speeds of train and light respectively. $\phi(x)$ is the angle between the base station and the train.

In practice, when an HST moves to the location x , the real Doppler shift it experiences on the LOS path $f_{(d,real)}(x)$, the current carrier frequency of LTE-R $f_{(carrier,train)}$, and its speed $v(x)$. Thus, $f_{(d,real)}(x)$ can be estimated from the data fetched from REM by [47]:

$$\hat{f}_{d,est}(x) = \left(\frac{f_{carrier,train} v_{train}(x) f_{d,REM}(x)}{f_{carrier,REM} v_{REM}(x)} \right) \quad (3.2)$$

$f_{(carrier,train)}$ is the known quantity, and to determine the other quantities for Doppler shift estimation, we require the knowledge of the train's location to estimate REM carrier frequency and velocity.

3.2 REM-based security and surveillance

Given the broadcast nature of wireless sensor networks in 5G and beyond communications networks, for example, in IoT-based applications, a need for surveillance and security to legitimate or licensed users in a given CR emerges. The congestion (available bandwidth to accommodate many users) of available spectrum in sub 6 GHz with densely deployed UEs raises the concern for severe interference between licensed and unlicensed users in a spectrum. Jammers, eavesdroppers, and exploratory attackers in a given geographic area are the main causes of interference apart from hidden node problems in CR enabled networks. In order to tackle these problems effectively and protect the primary transmitters in an area against fake base stations, whose main objective is to evade spectrum fees, a proactive solution is generating

REM for monitoring irregular activities. REM characterizes long-term spectral monitoring and monitors the legal activities and deviation of the radio map from normalcy or a predicted pattern. The radio map can help identify the location, PSD (spectrogram), and other important parameters of rogue transmitters. Legitimate communication can be switched to safer frequency bands to mitigate the effects of abnormal transmissions. On the other hand, the system operator can dispatch personnel to physically repair malfunctioning devices/BSs or capture the illegitimate transmitters [42].

The need for security against malicious entities in a wireless transmission scheme has always attracted constant attention. However, as a network's applications and reliability (as in military and IoT-based cases) increase in future and present communication systems, the requirement to evade such attacks increases. Building on the important aspect of security and 5G use cases, we briefly mention a very popular scenario of vehicle-to-everything (V2X) communication, which encompasses most of the 5G use cases and the need for REM to provide security and robustness to the applications.

3.3 REM for V2X and safety concerns in V2X applications

In recent years, in terms of research and implementation in the automotive market, the transport industry has witnessed tremendous growth in the evolution of vehicular communications technology, given the rise of intelligent transportation systems (ITSs). ITS relies on wireless communications and new computing and sensor technologies to provide more efficient, safe, and comfortable mobility in various vehicular scenarios. In particular, V2V and vehicle-to-infrastructure (V2I), generally encompassed under the term V2X, are fundamentally required for purposes such as the prevention of vehicle-to-vehicle collision, emergency notification to distant vehicles, route and track optimization, and automated driving [49]. One of the reasons V2X presents an exciting use case is because it combines the flavors of all 5G services. For instance, the collision avoidance messages between vehicles correspond to uRLLC use case due to their strict reliability and latency requirements; infotainment services fall under the eMBB umbrella due to high data rate requirements and high connectivity of massive machine-to-machine type connections (mMTC) [49].

This section highlights ITSs and V2X communication, exemplifying the need for a secure radio environment awareness and mapping process. From a financial standpoint, V2X networks present a massive opportunity in terms of income and research [49]. The numerous sensors onboard autonomous vehicles and their need to communicate with each other, road-side units (RSUs), and other traffic infrastructure necessitate an intelligent and secure integrated sensing and communication system to ensure a safe, economical, and more efficient driving experience.

The importance of this use case is also illustrated by the various standardization efforts carried out to fulfill its requirements. The two leading candidate families for V2X: IEEE 802.11p based dedicated short-range communications (DSRC) and ETSI's ITS-G5 standards, and the 3rd Generation Partnership Project (3GPP) backed cellular V2X (C-V2X) standard [50]. This distinction has led to various research on the coexistence of these two standards and how their

comparison can benefit the V2X use case. Bandwidth congestion implicitly affects these methods because of many sensors needed for reliability. Furthermore, autonomous vehicles would require much higher bandwidths for image processing applications, which these standards might not support. C-V2X, on the other hand, might be able to address these issues but it would take a while to ensure the provision of a communication infrastructure strong enough to support the requirements of V2X users. It is also possible that both standards would converge at some point; DSRC/ITS-G5, for instance, might be reserved for the V2V aspect of the communication in such networks [50].

ITS can support many useful applications by utilizing REM-based radio environment awareness and advanced computing, communication, and sensing technologies. These applications include vehicular and pedestrian safety, fully/semi-autonomous driving, remote driving, traffic flow optimization, and efficient routing. REM generally increases radio environment awareness and contains information that can be captured through external infrastructure and V2X communication standards. The overall REM database in such scenarios contains information about vehicle-based user equipment, RSU, signs, traffic lights, parking zones, smart barriers, and mobile infrastructure. The integration of a vehicle's communications with other vehicles (V2V), infrastructure (V2I), network (V2N), pedestrian (V2P), environment (V2S), ecosystem "V2E" and transport network (V2TN) [50].

Building or updating REM in V2X is vulnerable to exploratory (generalized eavesdropping through a sensor node), manipulative (meaning manipulation of the true REM with evasive false information which leads to premature REM updating according to the malicious intent of a node in the network), and destructive attacks (interference, blocking of signals through noise). These attacks can corrupt secure and sensitive information affecting the overall functionality of different ITS entities by attacking detection, communication, and control capabilities [50]. For example, these attacks can affect the environment detection, controllability (speed, direction, and braking system of semi/full vehicle, autonomous driving, and vehicle-related functions and the external environment [49]. It can cause cargo damage, endangering the lives of pedestrians, drivers, and passengers and impairing the overall effectiveness of the ITS. These attacks cause harm to the provided REM as an input to the autonomous driving vehicles, which incorporates these inputs to predict the propagation channels, messages like collision, and precise knowledge of the traffic. As a result of such malicious intent from nodes, REM manager tends to change the information regarding the environment, such as route details, blocking artifacts, and traffic in the route, to their advantage. These frequent updates in REM increase system complexity and thus raise security threats for long-term environment management.

Some attacks on REMs and their consequences for the respective ITS are described below.

3.3.1 Type of threats and solution

An important issue is the security and authenticity of REM and sensing information, which is easily susceptible to eavesdropping, spoofing, and jamming attacks. Similarly, the security and legitimacy of the sensing signal, along with the overall radio environment awareness and mapping process, are also susceptible to exploratory, manipulation, and disruption attacks. With the significance of wireless sensing evident for future wireless communication and other

applications, large-scale wireless sensing will only be feasible if methods are developed to prevent, detect, and negate these attacks. The threats can be categorized into three major types:

1. **Exploratory attacks:** This entails that attackers have access to the information about the vehicle, vehicle network, and the confidentiality of the REM. The information stored in the REM manager is prone to harmful entities. The attacker can access vital information about the environment, such as vehicle route, traffic in the area, power, engine, speed of the vehicles. Access to this information enables attackers to alter specific information, for example, the vehicle's location and places. Solutions to such attacks include cooperative jamming solutions, multihop V2X relaying, reconfigurable intelligent surfaces (RIS)-assisted solutions, and physical layer key extraction-based techniques [49].
2. **Manipulation attacks:** In this scenario, the attacker can manipulate vehicle and REM-related information such as false traffic routes, misleading obstacle information, incorrect information about other vehicles, etc. This can be achieved by sending or rebroadcasting false GPS information about speed, trajectory, and channel variations that cause the REM to update its stored information frequently and cause high disruption in environment estimation, which inherently leads to accidents or rerouting of the vehicle it attacks. Solutions to this problem include active and passive authentication schemes, distillation, crowdsourcing, and randomized probing in time/space-based techniques, which are explained in more detail in [49] and are out of the scope of this thesis.
3. **Disruption attacks:** Caused by employing jammers that hide in the network as normal nodes and cause interference in the REM maps. These interferers cause accidents by transmitting noise-like signals, which confuses the REM manager through unnecessary collision messages. Solutions to these attacks include cooperative relaying via V2X nodes, spread spectrum coding, machine learning-based solutions, and RIS-assisted anti-jamming [50].

A more detailed overview can be found in the reference [49].

3.4 REM assisted physical layer security

REM and sensing in V2X communication help provide cognitive physical layer security (PLS). The threats mentioned in the previous section affect the physical layer of a communication system majorly collision messages for V2X technology. Thus it becomes imperative to offer cognitive and robust solutions to the physical layer of the open system interconnection (OSI) model. Increasing the signal-to-noise interference level (SINR) of a particular system fails to address this problem as the jammer can improve its SINR through more number of antennas. Even the use of channel reciprocity also fails to provide a better solution provided the concept fails a certain situation [49].

However, the use of REM and sensing provides a more robust solution. REM consists of several physical layer communication characteristics such as modulation scheme used, waveform employed, and car properties such as engine, speed, location, and information about the environment. All these vital queries are acquired using sensing schemes and building up REMs. The information collected is fed to a risk identifier block that performs some signal processing

methods and identifies possible threats using historic environment information from the REM. The output of this risk identifier block is used to allocate resources effectively such as possible threats, jammers, or interferers in the cell. As a result, REM and sensing have grown from being an essential part of CR networks to finding their roots in future communications networks and various applications apart from spectrum sensing, interference, and power cartography.

3.5 REM construction for UAV-based application

The literature on environment map construction is focused mainly on power and interference map construction. Deploying 5G and 6G cellular communication technology, an upcoming application, UAV-assisted cellular communications, is gaining more interest from researchers. REM-assisted networks for CR have been extensively studied for 5G functionalities such as reliable communication, high volume of users in the allocated bandwidth providing higher data rates to users, and solving latency-related issues. However, REM finds its benefits in CR networks, robotic-based applications, and other useful scenarios, including IoT and surveillance.

Robotic-assisted communication includes the use of UAV as a promising technology for relay-assisted communication terminals. UAVs can act as aerial base stations to provide connectivity to users hindered by obstacles or regions where terrestrial connectivity from established BSs is hard to reach. The literature on cell-based communication is well-defined in the case of static base stations. However, UAVs act as helping hands to provide connectivity where the signal strength from these established networks is severely hindered by shadowing objects or are just out of coverage region.

UAVs are a promising aspect in scenarios such as concerts, festivals, or large-scale events where the allocated spectrum in a region becomes highly congested because of such a dense network of end users. Another scenario where the UAVs could be helpful is when disaster hits geographic regions where the established network infrastructure is highly compromised or damaged. Providing network connectivity to the users in such cases becomes important, for instance, rescue operations for first aid responders in a disaster hit area. UAVs can help establish network coverage for users on the fly with the assistance of REM maps, which provide online estimation of the channel and spectrum availability in the particular region of interest. Therefore, REM construction using measurements collected from the UAVs and deploying UAV as an aerial base station for severely hindered coverage regions is an upcoming and interesting research topic.

The challenge encountered in such an application is that the UAV user channel is unknown before sending the UAV to the target position. Therefore, learning the air-to-ground channel for every possible UAV-user location pair is essential. In other words REM construction in time, frequency, and spatial domain is of utmost importance. The UAV-assisted wireless channel is characterized as a deterministic function of transceiver distance pair as in RSS-based REM construction or a temporal and spatial correlated process from the prior measurements taken at nearby locations [51]. In the case of cellular network applications, the presence of the urban architecture or vegetation may critically affect the performance. A possible solution to the shadowing problem is to use a stochastic model to capture LoS links, which are more probable in the UAV-based applications (direct UAV to user pair). In [51], authors implement a ray

tracing-based channel estimation algorithm for channel gain map construction. The air-to-ground channel is reconstructed for every UAV-user location pair using direct path ray-tracing analysis based on the estimated multi-class virtual obstacle map and radio map reconstruction is reduced by 300 times [51]. In [52], active sensing-based dynamic REM estimation due to the changing trajectory of the UAV is studied to enhance the radio environment estimation problem.

An exemplary scenario of a UAV-based wireless communication system is shown in the figure.

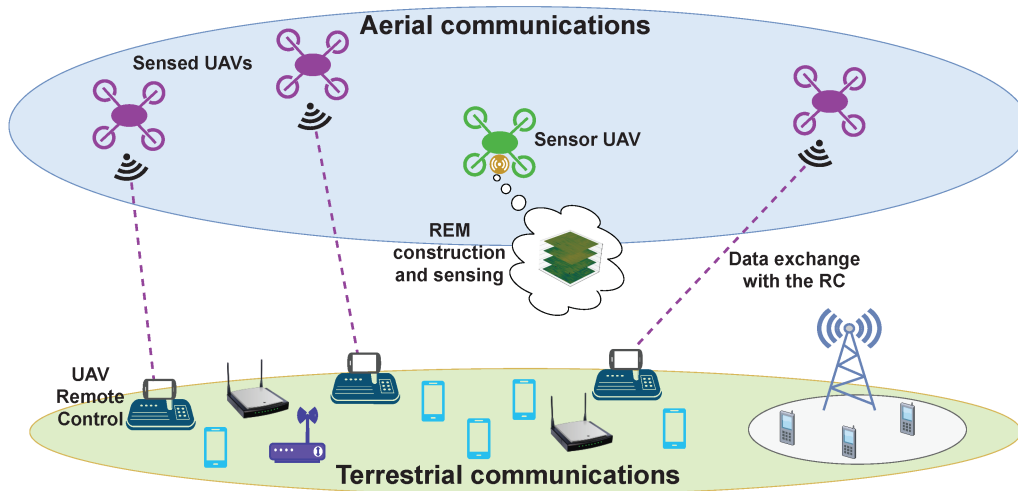


Figure 3.2: UAV-based communication scenario [53].

3.6 Chapter summary

This chapter discussed several state-of-the-art REM construction methods based on 5G use cases, such as high data rate, ultra low latency, high mobility, MIMO for high throughput, and reliable connection. We present a comprehensive and brief introduction to the REM construction and REM update rate through application-based scenarios in use cases such as V2X, UAV, and HST's. We present a brief overview of the underlying concepts that define each use case separately and analyze the REM construction method for a specific application and use case.

4 Methods

In the previous chapters, we have introduced the basics and fundamentals of WSNs, localization problems in WSNs and REM, REMs as a utility in CR networks, and REMs as an important tool in 5G and 6G cellular networks based positioning or IoT-based applications. This chapter presents the preliminaries of the system model required for implementing transmitter localization methods. We first discuss the RF propagation phenomena in free space (scattering, diffraction, path loss attenuation, and use of log-normal path loss for RSS measurements), further on to elaborate on the wireless propagation of RF signals we discuss fading (small-scale and large-scale) in wireless channels and present the case of received power in a dedicated sensor network in a given area of interest which will be elaborated in the later section separately. The system model comprises received power as a function of the path loss model and a dedicated sensor network arrangement (including mention of some standard sensors used in experiments) incorporating the knowledge of the known transmitter power for the localization problem. Additionally, we define the two localization methods with a detailed discussion of their mathematical formulation. After introducing the preliminaries, we present the system of dedicated sensor networks, making our arrangement of sensors in a grid-like structure evident. It also explains the scenario with an increasing number of deployed sensors for the performance evaluation. After the basics of the system model, we present the first results for evaluation. After establishing the groundwork for localization methods and their performance evaluation, we discuss OFDM signals and how their implications can impact the localization methods. This part of the thesis looks to incorporate the signals used for 5G NR use cases and bring the thesis relevance to the up-to-date technology deployment.

Finally, we study the correlated shadowing model for real-world scenarios and present the results for varying shadowing noise standard deviation cases.

4.1 RF propagation

The electromagnetic (EM) waves theory states that any acceleration of an electric charge or variation in electric and magnetic fields produces radiation. They are a superposition of electrical and magnetic waves traveling through the free space as shown in Figure 4.1. The high-frequency current through the antennas of a transmitter produces these EM waves. These waves exhibit properties of light and oscillate in the free space with the speed of light ($c = 3 \times 10^8 \frac{m}{sec}$). The direction of propagation of the electric field and the magnetic field is perpendicular to each other and the vector representation of the EM wave is given by Maxwell's equation:

$$\vec{S} = \frac{1}{\mu} (\vec{E} \times \vec{B}) \quad (4.1)$$

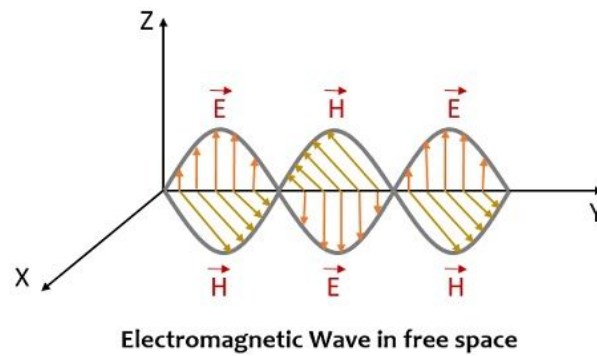


Figure 4.1: Electromagnetic wave propagation [54].

In the equation, μ is the free space propagation constant, $\vec{E}$ and $\vec{B}$ are the electrical and magnetic fields respectively, and $\vec{S}$ is the EM wave. Radio propagation is how radio waves are transmitted from one end to another inside the earth's atmosphere. Figure 4.2 shows the phenomena affecting these radio waves in free space. The waves transmitted from one point inside the Earth's atmosphere to another are affected by reflection, scattering, and diffraction. The following subsections briefly discuss these phenomena starting with the free space propagation model.

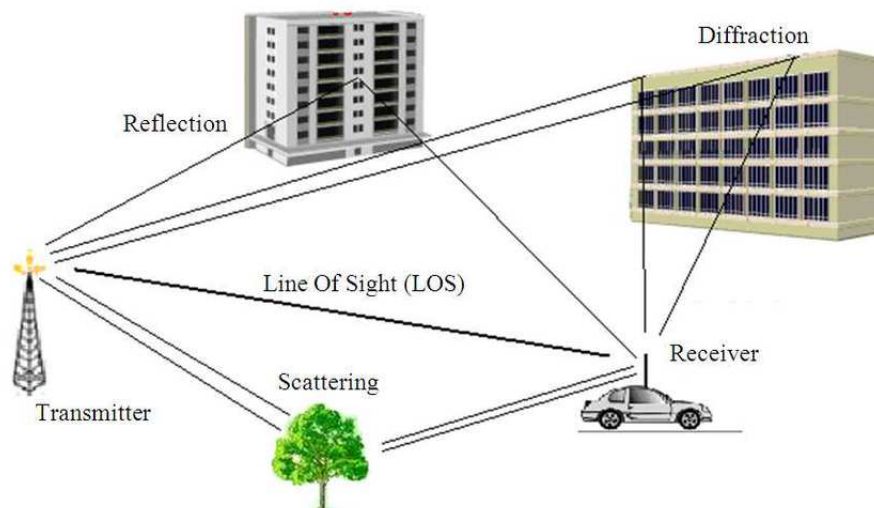


Figure 4.2: EM waves affected by scattering, reflection, and diffraction in free space propagation model [29].

4.1.1 Free space propagation model

The free space propagation model predicts the power at the receiver's side. This model is characterized by the presence of an unobstructed direct or straight line path (LoS) between the transmitter and receiver with no multipath components. The signal energy is radiated in an omnidirectional manner from the aperture of the antenna, and it behaves as a point source. Assuming there is no energy loss, the free space model helps predict the attenuation at the receiver side as the transmitted signal passes through obstructions. The power received at the

receiver antenna as a function of the distance between the Tx and the Rx is given by Friis's equation:

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi d)^2} \quad (4.2)$$

where the parameters used are defined as:

- $P_r(d)$ is the received power as a function of distance (d) between Tx and Rx.
- P_t is the transmitter power.
- G_t and G_r are the effective antenna gain factors at Tx and Rx.
- λ is the wavelength of the signal.

The antenna gain G is given as:

$$G = \frac{4\pi A_{eff}}{\lambda^2} \quad (4.3)$$

where A_{eff} is the effective aperture of the antenna. The path loss or attenuation is defined as a function of the distance between the Tx and Rx, frequency of operation, and antenna gains as:

$$PL(d, \lambda, G) = \frac{P_t}{P_r} = \frac{(4\pi d)^2}{\lambda^2 G_t G_r} \quad (4.4)$$

The received signal power in decibels as a function of transmit power, path loss, and replacing wavelength in terms of frequency ($\lambda = \frac{c}{f}$) is written as:

$$P_r(dB) = P_t + G_r + G_t - (32.44 + 20 \log d + 20 \log f) \quad (4.5)$$

$$= P_t + G_r + G_t - PL \quad (4.6)$$

4.1.2 Reflection

EM waves behave as light waves and show similar effects when they encounter a surface with dimensions similar to the wavelength of the wave. When a radio wave travels with the speed of light in the terrestrial plane and hits an object with a comparable wavelength it suffers two phenomena reflection and refraction. We focus our interactions in the same medium and hence only refer to the reflection of the signal. The reflected wave follows Snell's law of reflection and reaches the receiver antenna with a certain angle of reflection and polarization. The reflection of the EM wave affects the RSS measurements at the receiver side which are of major importance. Reflected transmitted signals are the main cause of multipath signals which add constructively or destructively depending on two quantities namely the time delay of arrival and the relative phase of each signal component. The reflection of a transmit signal is depicted in Figure 4.2.

4.1.3 Diffraction

Consider a scenario where a transmitted wave follows Huygens's principle of spherical wavefront propagation. The propagated wave has spherical wavefronts as it is closer to the source and becomes planar as it moves far away from the source characterizing far field and near field. When this propagating wave encounters an obstructing object, for example, the edge of an object the propagating wave bends around the object while the object blocks the rest of the wavefront. As it hits the object's edge, it appears a new wavefront propagates beyond that point. This phenomenon is termed diffraction and is prominent in shadowing objects.

4.1.4 Scattering

Another aspect of wireless transmission is scattering. As the transmitted wave hits a surface which is not smooth as in the case of reflection, the propagating waves scatter or disperse in the medium. The dimensions of the scatterer are smaller as compared to the wave. In wireless transmission, the communication network consists of several scenarios that affect the received power causing attenuation of the signals. Scattering is especially prominent in urban or semi-urban scenarios where the signal encounters shadowing objects such as dust particles and leaves or any object with a size smaller than the wavelength of the impinging wave.

4.2 Fading

Fading in wireless communication is a characteristic that affects the amplitude of a transmitted signal, especially its effect can be seen on the RSSI measurements at the receiver side. Fading in wireless communication is the property of the wireless channel where the signal amplitude corresponds to variations in time and frequency. Fading (non-additive) is another common phenomenon impacting signal degradation besides additive Gaussian noise. Fading is categorized into two categories *small scale* fading and *large scale* fading depending on how severely the signal deteriorates given the environment and channel characteristics. The impact of fading on RSSI measurements is of importance in modeling the path loss model. Different and varying scenarios require an appropriate choice of the path loss models; hence, having an acute understanding of the channel model based on application should not be overlooked. Classification of fading channels is shown in figure 4.3.

In the following sub-section, we categorize small-scale and large-scale fading and determine the path loss models and how they affect the received signal.

4.2.1 RSS measurements in large-scale fading channel

Large-scale fading or slow fading occurs when a mobile user moves through large distances of the size of the cell. The signal attenuates because of the presence of shadowing objects such as buildings, terrain, and vegetation. Slow fading also arises when the coherence time of the channel is large relative to the delay spread of the channel. The slow fading is characterized by average path loss and shadowing.

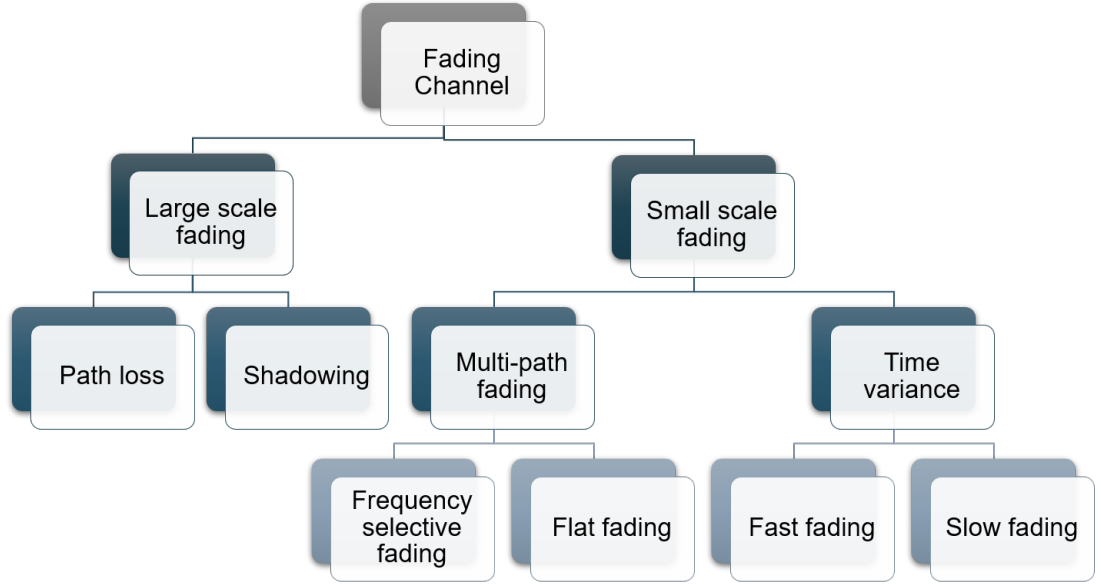


Figure 4.3: Fading classification [2].

Shadowing is defined as the variation of median path loss as the signal travels through and around large objects or obstructions. The signal amplitude in shadowing is modeled using log-normal distribution. The path loss model for slow fading and RSS measurements are calculated in decibels and this path loss model is a simple case of free space path loss propagation model called as log distance model. The shadowing variation follows a normal distribution, making it easier to model the received power levels in the dB scale. The RSS measurement at the i -th sensing unit denoted by P_i^{Rx} is measured as:

$$P_i^{Rx}(dB) = P^{Tx}[dB] - 10\alpha \log \left(\frac{d_{(x_i, y_i)}}{d_0} \right) + S_i \quad (4.7)$$

Where P^{Tx} is the transmit power, α is the path loss exponent, $d_{(x_i, y_i)}$ is the distance between the primary transmitter and the sensing units and d_0 is the reference distance. S_i is the shadowing Gaussian random variable with distribution $S_i \sim \mathcal{N}(0, \sigma_S^2)$.

The RSS measurements may vary significantly because of the variance of the shadowing distribution. Therefore, to mitigate the impact of shadowing, we average the RSS measurements over the number of samples:

$$\bar{P}_i^{Rx}(dB) = \sum_{j=1}^{N_m} \left(\frac{P_{i,j}^{Rx}}{N_m} \right) \quad (4.8)$$

where $\bar{P}_i^{Rx}$ is the average RSS value at i -th node and N_m are the number of samples.

4.2.2 RSS measurements in fast fading channel

Fast fading or small-scale fading occurs because of the multi-path or reflected signals that add up constructively or destructively at the receiver antenna. Fast fading is characterized when the coherence time of the channel is small as compared to the max delay spread (bandwidth) of the channel. As the mobile user moves shorter distances the delay of the multipath signals causes the change in doppler or frequency shift and thus the coherence time of the user channel. The signal amplitude varies rapidly around the mean path loss curve. Here instead of the log-normal model, we employ the use of Rayleigh fading to account for multipath signals arising from various scatterers and diffusers in the propagating environment. The probability density function of the Rayleigh fading channel is given by:

$$f(\gamma) = \frac{1}{\bar{\gamma}} \exp\left(\frac{-\gamma}{\bar{\gamma}}\right) \quad (4.9)$$

where $\bar{\gamma}$ is the average SNR of the distribution. The received power in this scenario follows the exponential distribution as opposed to log-normal distribution given by [9]:

$$P_i^{Rx} (Watts) = E_i(10^{\frac{\phi}{10}}) \quad (4.10)$$

Where ϕ is given as an exponential term:

$$\phi = P^{Tx} - 10\alpha \log 10\left(\frac{d_{(x_i, y_i)}}{d_0}\right) + S_i \quad (4.11)$$

Since Rayleigh and shadowing cause statistically independent fluctuations over the channel, the shadowing term can be added directly to the exponential distribution of fast fading channel.

4.3 RSS-based localization methods

The following sections and subsections discuss the system model in detail. Firstly, we detail the two localization methods and their mathematical formulations. Secondly, we introduce the sensor network design and present the scenario with sensor nodes and active transmitters in a grid-like structure. After presenting the methods and sensor network design we present the simulation results.

4.3.1 MSE-based method

After gaining an insight into the path loss model and fading, we shift our focus to the transmitter localization methods. We begin our detailed discussion on these methods starting with the MSE-based method as described in [6]. As mentioned in equation 4.7, the received power at the i -th node in the network is given by the equation:

$$P_i^{Rx}(dB) = P^{Tx}(dB) + K(dB) - 10\alpha \log_{10} \left(\frac{\sqrt{(x_i - x_{pu})^2 + (y_i - y_{pu})^2}}{d_0} \right) \quad (4.12)$$

In the equation 4.12, K is the unitless constant that depends on the antenna characteristics of the transmitter and the receiver and contains the frequency components, (x_{pu}, y_{pu}) are the true transmitter coordinates to capture RSS values for simulation. We rearrange and rewrite the equation (x-above equation) as follows:

$$x_i^2 + y_i^2 = 2x_{pu}x_i + 2y_{pu}y_i + d_0^2 10^{\left(\frac{P^{Tx} - P_i^{Rx}}{5\alpha}\right)} - R^2 \quad (4.13)$$

Again after some manipulations and rearranging the terms in the above equation, we determine the above problem in a linear matrix form:

$$A = \begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) & 10^{\frac{-P_{r2}}{5\alpha}} - 10^{\frac{-P_{r1}}{5\alpha}} \\ 2(x_3 - x_1) & 2(y_3 - y_1) & 10^{\frac{-P_{r3}}{5\alpha}} - 10^{\frac{-P_{r1}}{5\alpha}} \\ 2(x_4 - x_1) & 2(y_4 - y_1) & 10^{\frac{-P_{r4}}{5\alpha}} - 10^{\frac{-P_{r1}}{5\alpha}} \end{bmatrix} \quad (4.14)$$

$$\theta = \begin{bmatrix} x_t \\ y_t \\ d_0^2 10^{\frac{PL_0}{5\alpha}} \end{bmatrix}; Y = \begin{bmatrix} x_2^2 + y_2^2 - x_1^2 - y_1^2 \\ x_3^2 + y_3^2 - x_1^2 - y_1^2 \\ x_4^2 + y_4^2 - x_1^2 - y_1^2 \end{bmatrix} \quad (4.15)$$

Equations 4.14 form a linear regression problem that could be solved in the mean squared error sense employing standard least squares or MSE solutions. The above matrix is of the form $\mathbf{A}\theta = \mathbf{Y}$, where $\mathbf{A}$ is the input parameter matrix containing the coordinates of the sensor nodes and the received power levels of each node. The variable is the unknown parameter that needs to be determined and it consists of estimated transmitter locations and reference path loss PL_0 . $\mathbf{Y}$ is the output parameter vector consisting of the squared difference between the transmitter and receiver coordinates. Both $\mathbf{A}$, $\mathbf{Y}$ are determined matrix and vector.

Working of algorithm: Any RSS-based localization scheme is based on multilateration i.e., involving three or more sensors for carrying out the measurements. Assuming the network has one common random access channel to the fusion center, only one sensor can send its measurements to the fusion center in each time slot. So, the localization begins only after the first four measurements have been recorded. After n time slots, we have a total number of $\binom{n}{4}$ combinations of sensor data to manipulate input system matrices $\mathbf{A}$.

The MSE problem is formulated as follows [6]:

$$J(\theta) = (A_1\theta - Y_1)^2 + \dots + (A_n\theta - Y_n)^2 \quad (4.16)$$

$$= \sum_{i=1}^n (A_i\theta - Y_i)^2 \quad (4.17)$$

To solve for θ in the MSE sense we differentiate the cost function with respect to theta and the solution is given as:

$$\frac{\partial J(\theta)}{\partial \theta} = \sum_{i=1}^n (2A_i^T(A_i\theta - Y_i)) = 0 \quad (4.18)$$

$$\theta_{opt} = \frac{\sum A_i^T Y_i}{\sum A_i^T A_i} \quad (4.19)$$

As a result of such combinations which could be in the order of thousands or more depending on the number of sensors used to capture the measurements the complexity of the algorithm increases. Since selecting the sensors and their corresponding power levels is purely random, we average the estimated location parameter for 20 experiments. The randomness of the node selection and the shadowing noise impact the localization problem which will be clear in the simulation results section.

The linear regression problem mentioned in the MSE-based method can be used to estimate the unknown parameter using the adaptive iterative method as stated in [6]. In this method, the linear regression method is expressed as:

$$\mathbf{Y} = \mathbf{A} + E \quad (4.20)$$

Where E is the instantaneous error and an initial guess for θ is provided to start with the gradient descent method as explained using the equation:

$$\theta_{i+1} = \theta_i + \mu A^T e_i \quad (4.21)$$

where e_i is the instantaneous error vector $e = Y - A\theta$ and μ is the step size. A major drawback of this method is that it requires measurements from all the sensors for optimization and estimation. However, the authors did not mention the exact system parameters they used and the corresponding references to the mathematical problem formulation and employed system parameters, which made it unrealistic to implement in a simulation environment. Therefore, the work refrains from further discussion over the method.

4.3.2 LlvE-based method

In [9], the authors incorporate the use of a path loss model and use it to produce a REM interference map for each frequency in use. As mentioned in Chapter two, RSS provides benefits

in terms of synchronization of transmitter and receiver and use of less hardware equipment than AoA and ToA based localization methods. Another advantage of using RSS-based schemes is the prior knowledge of transmitter power for localization.

To estimate the active transmitter position we employ the use of the equation (from MSE model) and rearrange it into a least squares regression problem ($\mathbf{A}\theta = \mathbf{Y}$) in terms of the input system matrix, output system matrix, and unknown parameter to be estimated as $\mathbf{A}$, $\mathbf{Y}$, respectively.

$$\mathbf{A} = \begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) & 10^{\frac{-P_{r2}}{5\alpha}} - 10^{\frac{-P_{r1}}{5\alpha}} \\ 2(x_3 - x_1) & 2(y_3 - y_1) & 10^{\frac{-P_{r3}}{5\alpha}} - 10^{\frac{-P_{r1}}{5\alpha}} \\ 2(x_4 - x_1) & 2(y_4 - y_1) & 10^{\frac{-P_{r4}}{5\alpha}} - 10^{\frac{-P_{r1}}{5\alpha}} \end{bmatrix} \quad (4.22)$$

$$\theta = \begin{bmatrix} x_t \\ y_t \\ d_0^2 10^{\frac{PL_0}{5\alpha}} \end{bmatrix}; Y = \begin{bmatrix} x_2^2 + y_2^2 - x_1^2 - y_1^2 \\ x_3^2 + y_3^2 - x_1^2 - y_1^2 \\ x_4^2 + y_4^2 - x_1^2 - y_1^2 \end{bmatrix} \quad (4.23)$$

We need at least four sensor nodes to solve for unknown parameter θ . The optimal value of θ can be formulated as a constraint optimization as:

$$\hat{\theta} = \text{argmin} \|\mathbf{A}\theta - \mathbf{Y}\| \quad (4.24)$$

$$\text{s.t. } \mathbf{r}^T \theta + \theta^T \mathbf{P} \theta = 0 \quad (4.25)$$

Where P and r are given by:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; r = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \quad (4.26)$$

Working of algorithm: To solve for the unknown parameter, we define the constrained optimization problem using the Lagrange dual function and then solve for unknown θ .

$$L(\theta, \lambda) = (\mathbf{A}\theta - \mathbf{Y})^T (\mathbf{A}\theta - \mathbf{Y}) + \lambda (\mathbf{r}^T \theta + \theta^T \mathbf{P} \theta) \quad (4.27)$$

where λ is the lagrange multiplier. Differentiating the Lagrange dual problem with respect to θ and λ the Lagrange decomposes into two differential equations.

$$\frac{\partial L(\theta, \lambda)}{\partial \theta} = 2\theta^T (\mathbf{A}^T \mathbf{A} + \lambda \mathbf{P}) - 2\mathbf{Y}^T \mathbf{A} + \lambda \mathbf{r}^T \quad (4.28)$$

$$\frac{\partial L(\theta, \lambda)}{\partial \lambda} = \mathbf{r}^T \theta + \theta^T \mathbf{P} \theta \quad (4.29)$$

Equating both the equations to zero and solving for equation (first from above) leads to θ to be substituted in equation (next one) as:

$$\theta = (A^T A + \lambda P)^{-1} (A^T Y - \frac{\lambda}{2} r) \quad (4.30)$$

We solve for the unknown parameter θ using numerical methods function in MATLAB called `fmincon`, which takes in the constraint problem and gives out the estimated problem. We mention the performance of the method in the results section.

4.4 Sensor network design

An important point to consider in every network design is the functionality of the components involved and the selection and placement of the hardware components. Therefore, it is essential to carefully think about hardware component selection and their placement in the network as the signals might be impacted by the wireless channel fading, density of the sensor network, sensor sensitivity, and synchronization with the fusion center and other sensors in the network to avoid interference.

As in the case of dedicated sensor networks, the distribution of sensors to perform distributed spectrum sensing in a given space is based on sampling methodology. Sampling corresponds to the fact that the arrangement of the deployed sensing units or sensor nodes will probably appear in an experimental setup, for example in the case of UAV-based applications for radio environment mapping. In that case, deploying three drones over the air and regions of interest and collecting power measurements helps in localization and power map construction using trilateration methods. However, in our problem statement making radio environment mapping simpler and constructing only power maps through transmitter localization schemes suffices our thesis goal.

As mentioned earlier, the choice of an appropriate sampling scheme is application-specific. The shape and variance of the area of interest and knowledge about the processes being modeled, for example, path loss modeling, must be considered when selecting a sampling strategy. Classic spatial sampling schemes can be subdivided into [16]:

1. Simple random sampling (SRS): selecting points to be distributed randomly.
2. Systematic sampling: either can be grid-like sampling or uniform spacing of nodes.
3. Stratified: Some regions are sampled more heavily than others.
4. hybrid approach

Two important properties to consider while sampling are: first, no two points are placed far apart. This decreases interpolation resolution. Secondly, measurements must be taken at sufficient distances and time stamps for less interference. In particular, clustered measurements are generally required to model small-scale effects and far-separated measurements for large-scale fading. Some typical hardware spectrum analyzers that can perform distributed spectrum sensing

used in the literature conducting experimental studies are USRP1, USRP2, and 2.4 GHz Atmel ATmega1281 microcontroller.

In the case of the transmitter power source, some typical examples of technologies used in the literature are WiFi, Zigbee, Bluetooth, and UWB for transmitting RSS power measurements. In our scenario, we sample sensing units using uniform sampling or spacing between the points. We consider the case where the deployment area is of size $100m \times 100m$ and we place the sensors in the given area as discrete points in a grid-like structure. We start the placement with 4 sensors and move forward until 49 sensors in total.

The following figure 4.8 shows the sampling in our grid-like structure where the blue dots represents the sensing units and a red dot represents the active transmitter.

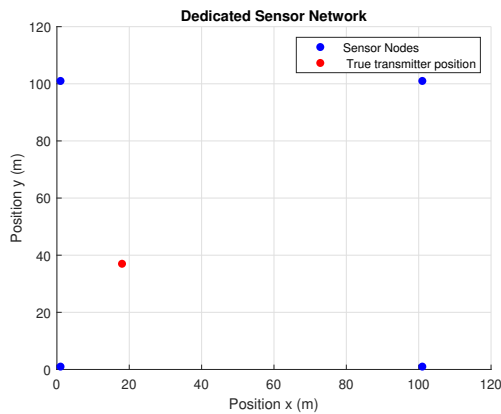


Figure 4.4: 4 sensors

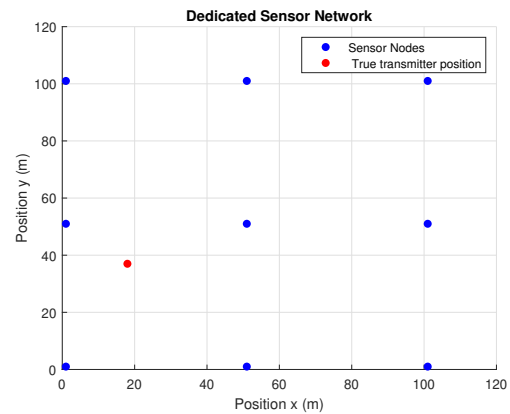


Figure 4.5: 9 sensors

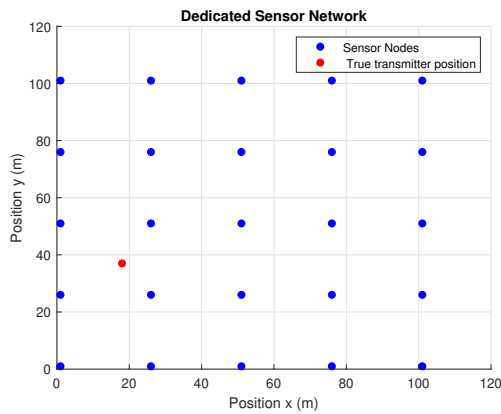


Figure 4.6: 25 sensors

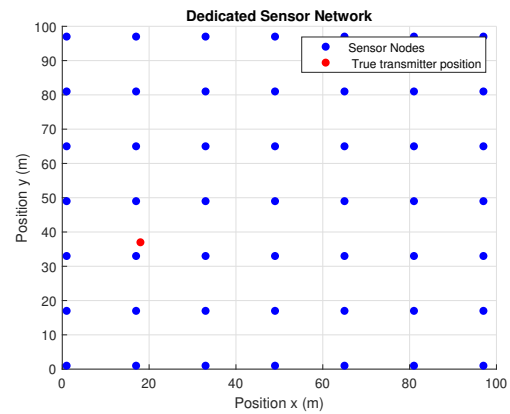


Figure 4.7: 49 sensors

Figure 4.8: Sensors placed in a grid-like structure.

5 Results

5.1 Simulation results

After describing the preliminaries for the system model, mathematical formulations for the two localization methods, selecting appropriate path loss model, and sensor nodes arrangement, we present the results comparing the performances of the two methods defined in the system model. The performance metrics we use for comparative analysis are the number of sensors in a pre-defined area and representing localization error vs number of sensor nodes. We also analyze the complexity, and accuracy of localization methods, the total number of required sensors or density, and the separation between sensors within the grid. These factors essentially impact the localization accuracy of the described methods and other RSS-based methods (like kriging, and Kalman filter - their variants, ML-based approaches) in general.

For our system model, we define the path loss model as in an urban scenario [6], [9]. The path loss exponent, log-normal shadowing standard variation σ_s , transmit power P^{Tx} , carrier frequency of operation f_c are given as 3.5, 8 dB, 0.1 Watts, and 3.7 GHz respectively with reference distance for calculating reference path loss as $d_0 = 2\text{m}$. The antenna gains G_t and G_r are considered to be 1 for simplicity. The scenario consists of one active and static transmitter at the position of (18m, 37m) and N varying sensing units that capture the RSS measurements. The point to note is that the considered scenario is static, meaning no additional movement or mobility of any of the stated components, thus establishing long-term fading, which we model as shadowing. We also follow the same system parameters for comparative analysis later on for the digital signal processing chain employing the use of OFDM for localization.

5.1.1 MSE-based method

In the following subsection, we present the pseudo-code for the MSE-based approach and then discuss the method's performance in terms of localization error vs. number of sensors. The results follow the system parameters mentioned in the previous section 5.1. Further, we present plots consisting of true REM and reconstructed REM using the estimated Tx positions from the unknown parameter θ .

In figure 5.1, we witness the performance of the MSE-based method against the number of sensors. The trend we infer from the figure is an increase in localization accuracy as the number of sensors to support localization increases. We also infer that the accuracy doesn't improve further or the performance converges as we keep incrementing the number of sensors from 25. This explains that in a small area of 100m x 100m, we don't see much improvement as we keep increasing the sensors. The impact of log-normal shadowing over the RSS variations

Algorithm 1 Location estimation using MSE method

```

1: Input 1: gridSize = (100m, 100m)
2: Input 2: numSensors = [4,9, 16, 25, 36, 49]
3: Calculate RSS measurements and average for 100 samples.
4: for num=1 to numSensors do
5:   Function call to calculate the transmitter positions as a function of (Tx, RSS, Number
     of experiments, and path loss exp).
6:   for iter = 1 to Numexps do
7:     Select Nsamples using nchoose(m,k) for computing combinations of RSS
       measurements.
8:     for i = 1 to Nsamples do
9:       Calculate system parameter matrix A and system output vector Y.
10:      unknown parameter =

```

$$\hat{\theta}_{opt} = (\sum A_i^T A_i)^{-1} (\sum A_i^T Y_i) \quad (5.1)$$

```

11:    end for
12:    Average  $\theta_{samples}$ 
13:  end for
14: end for
15: Output: Vector  $\theta_{opt}$  for numSensors.

```

is shown at the point in the figure with 25 sensors when a sudden jump causes localization accuracy to drop. We get results as close to less than 5 meters accuracy in terms of position error from the transmitter's true positions, which is not accurate as claimed by the authors in [6]. A limitation of this implementation is the density and area of interest, i.e., when the area is small for a given number of sensors (36), we see a trend of decreasing localization error against an increasing number of sensors. However, for the particular fixed scenario, if we change the radius of localization, it incurs an increase in the area of interest which in turn means the lower density of sensors in the area as depicted in figure 5.2 or other words greater separation between the sensors and between sensors and Tx. This causes the localization error to increase because the separation between sensor nodes and sensor nodes to the transmitter is quite large as opposed to previously. Therefore, for accurate localization, it is highly important to consider the area and sampling (separation between sensors) between sensors, and sensors should be placed near the Tx.

5.1.2 LlvE-based method

The parameters used for carrying out the simulation results are the same as in section 5.1. The results of the localization error are averaged for 100 samples in calculating the RSS measurements and over 25 experiments to reduce the impact caused by high shadowing noise. We average the measurements to account for variations in the environment for large-scale fading. This provides us with enough measurements to account for changing environmental conditions and to know how much accuracy we lose. Figure 5.3 describes the simulation results of the LlvE

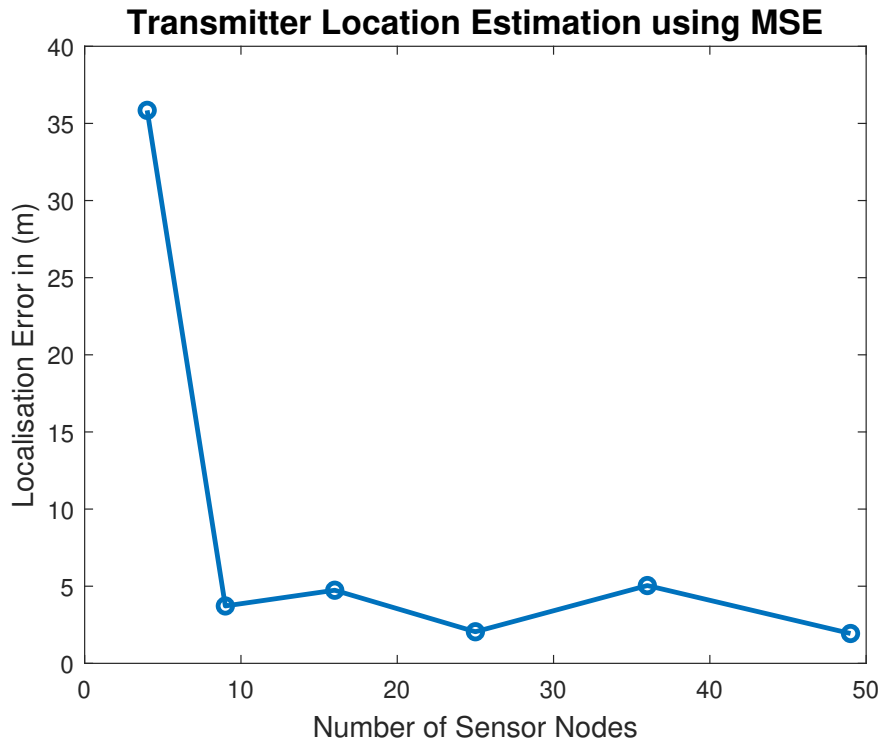


Figure 5.1: MSE-based localization method accuracy against the number of sensors deployed in the area.

algorithm's performance for log-normal shadowing and later in section 5.3, we present the case for localization performance based on correlated shadowing.

In figure 5.3, the localization error for the LIE method for the case of log-normal shadowing. The simulation result follows the same trend as in the case of MSE-based method. The performance of the two methods is quite similar as the methods are similarly defined from a mathematical point of view. The major difference that leads to slightly variable performance is because of the difference in implementation. In the case of MSE, we employ the use of only four sensors at a time to formulate the input parameter and output parameter matrix and vector, respectively, and use all possible combinations to derive the MSE solution and thus calculate the unknown parameter θ . However, in the case of implementing LIE method as opposed to MSE approach, we employ the use of all the sensor measurements available at a time and formulate the input and output for the linear regression problem along with non-linear constraints as defined in the section 4.3.2.

Using these non-linear constraints and RSS measurements of every available sensor explains the slightly better localization accuracy achieved through the LIE method. The accumulating data from every sensor and also optimizing the non-convex optimization problem in section 4.3.2 gives rise to a major limitation of the LIE method, which is the complexity in the run time of the simulation and infeasible points for the optimization problem which manifests as an uncertainty in the localization performance as compared to the linear solution and better run time complexity of the MSE approach. While finding a feasible point for the nonlinear constraints through the MATLAB debugger and 'fmincon' function for varying sensor network

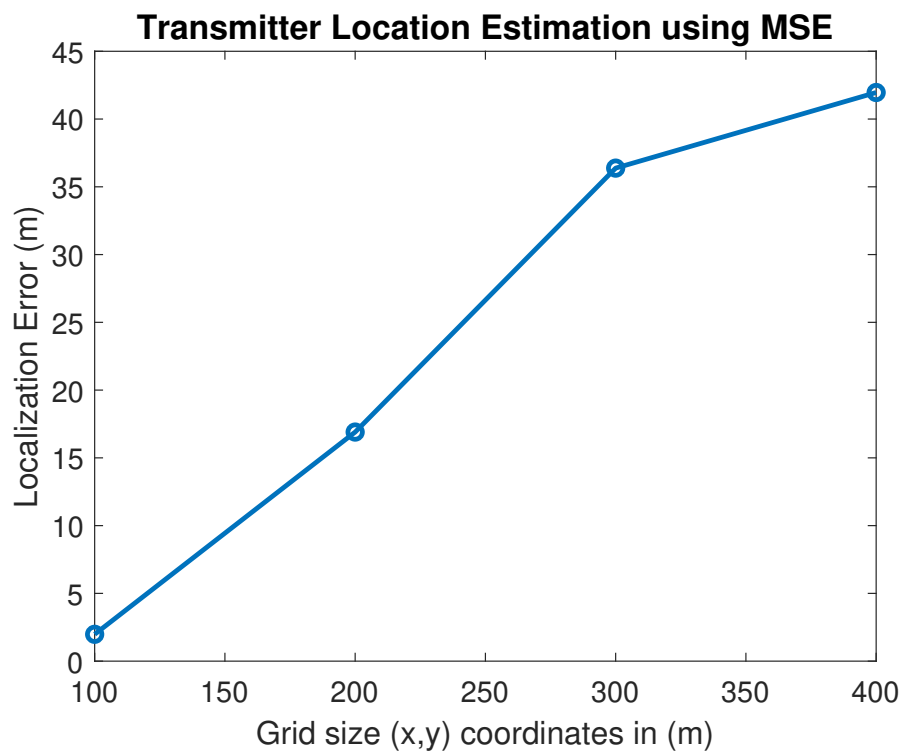


Figure 5.2: Localization error against the area of interest with 36 sensors.

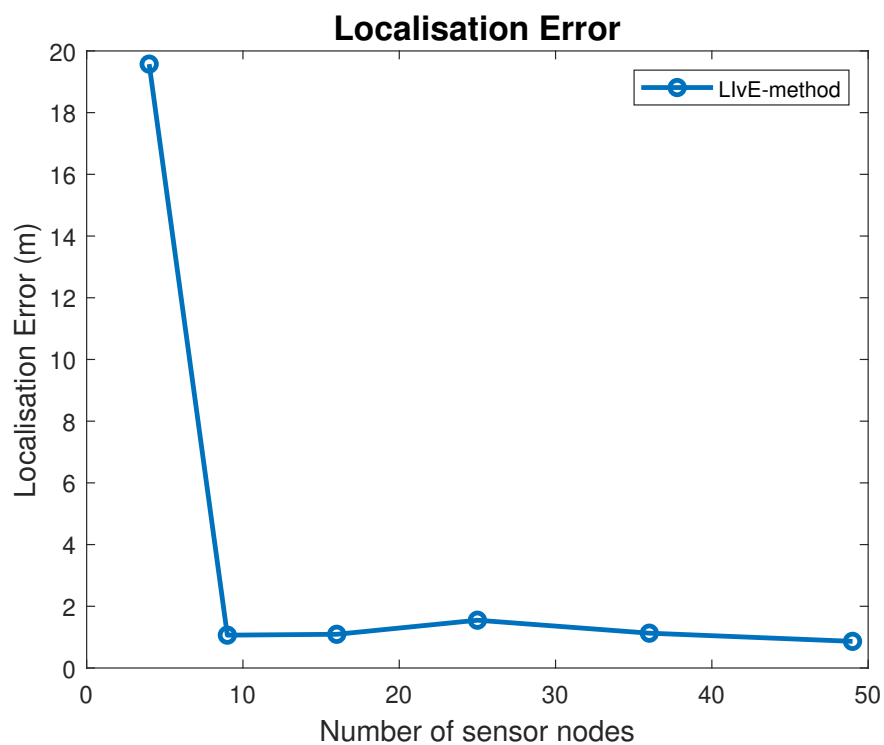


Figure 5.3: Localization error for LlvE method in meters w.r.t Tx position (18m, 37m).

arrangement, the function fails to find a convergence point for the optimization constraints. This means the optimization constraints are poorly defined, leading to inaccurate results. Therefore, for REM reconstruction and for further changing our transmit signal, we only limit our use to a based approach.

5.2 MSE and LlvE performance evaluation

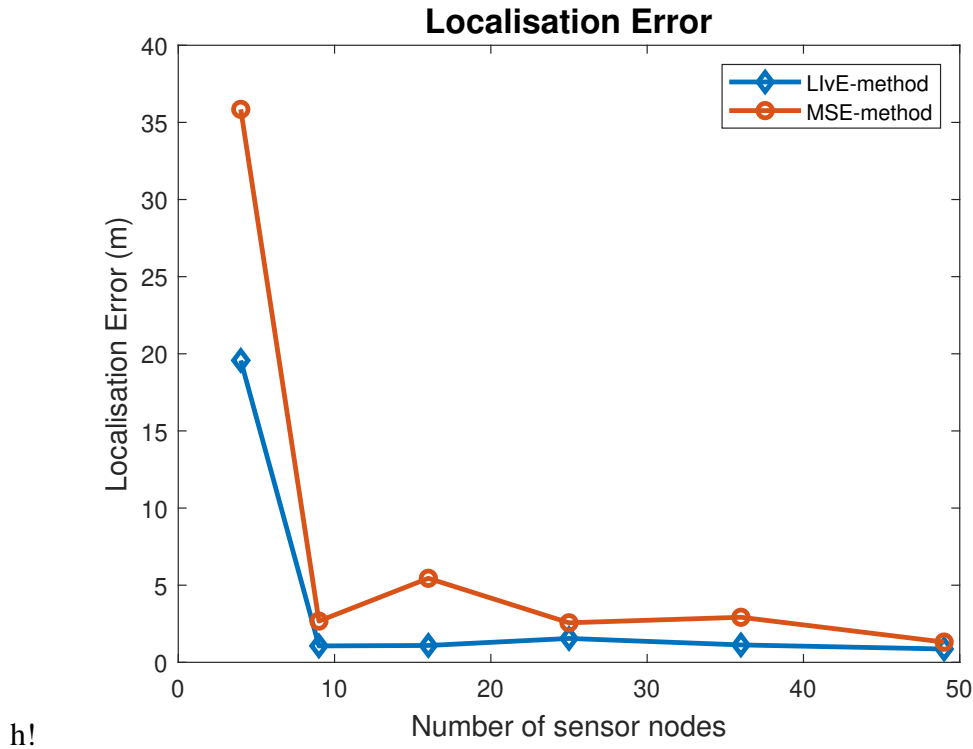
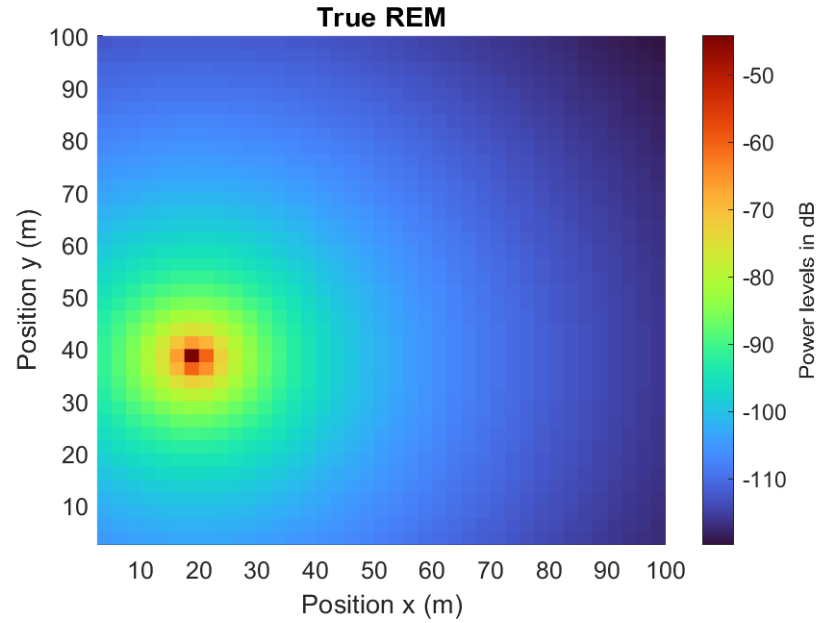


Figure 5.4: Localization performance comparison between LlvE and MSE methods.

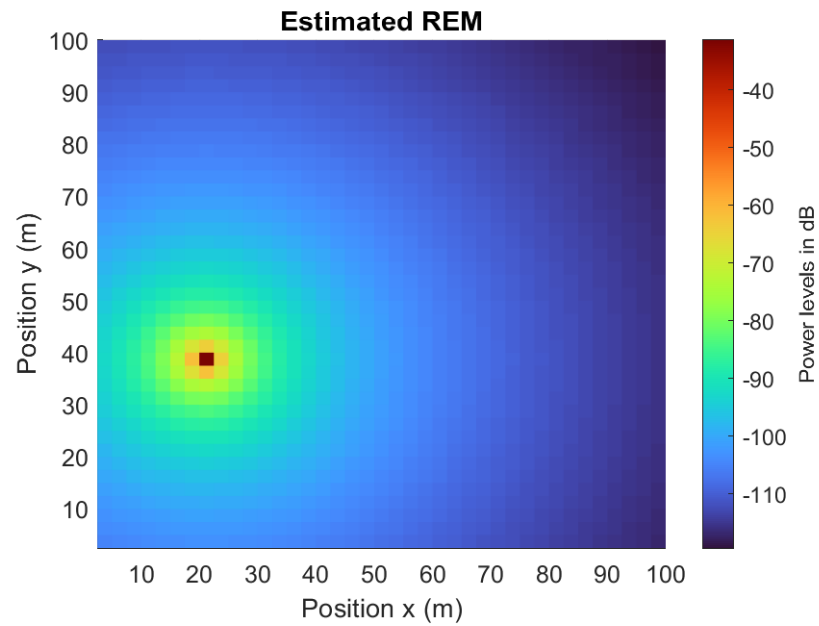
The simulation result, as shown in figure 5.4, clearly the LlvE-based approach performs better in terms of localization accuracy for given system parameters. There seems to be a significant difference in terms of localization accuracy until the number of sensors reaches 25. Adding more than 25 sensors, we see both methods converging to a similar value and don't witness much improvement as the number of sensors increases after 25. However, the run time complexity of the LlvE is much larger than the MSE, which gives us the tradeoff of better localization over run time complexity caused by solving non-linear constraints. Therefore, we shift our focus over MSE-based approach for REM construction and in the section 5.3.

In figures 5.5a and 5.5b, we present the comparison of the true and estimated REM after transmitter localization using MSE method. As seen in the figures, the estimated REM is as close to the true REM with some artifacts (similar red spots or heat spots) suggesting other possible transmitter locations with quite close power levels. The area in the heat plot with blue spots refers to the area with lower power measurements. The two simultaneous plots are, in fact, without shadowing noise, and because of that, we don't see much variation in power levels,

which is incurred by RSS variations caused by shadowing noise. Figure 5.6 shows an alternative view of the true REM where the peak of the curve represents the highest RSS power at the true transmitter position.



(a) True REM.

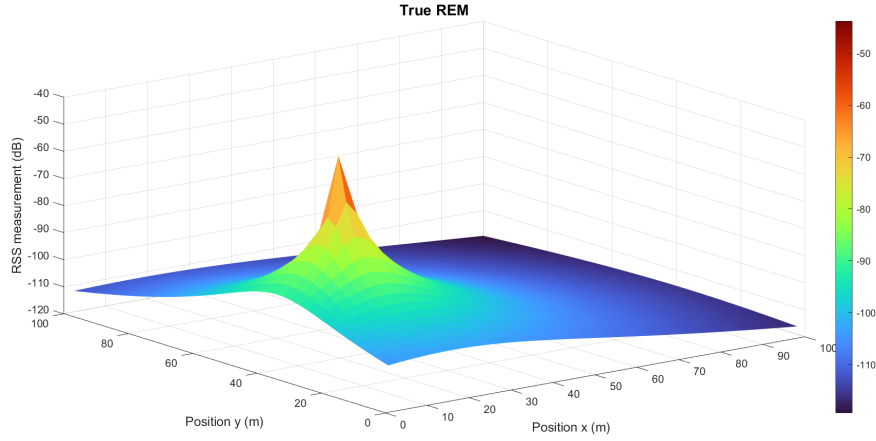


(b) Estimated REM.

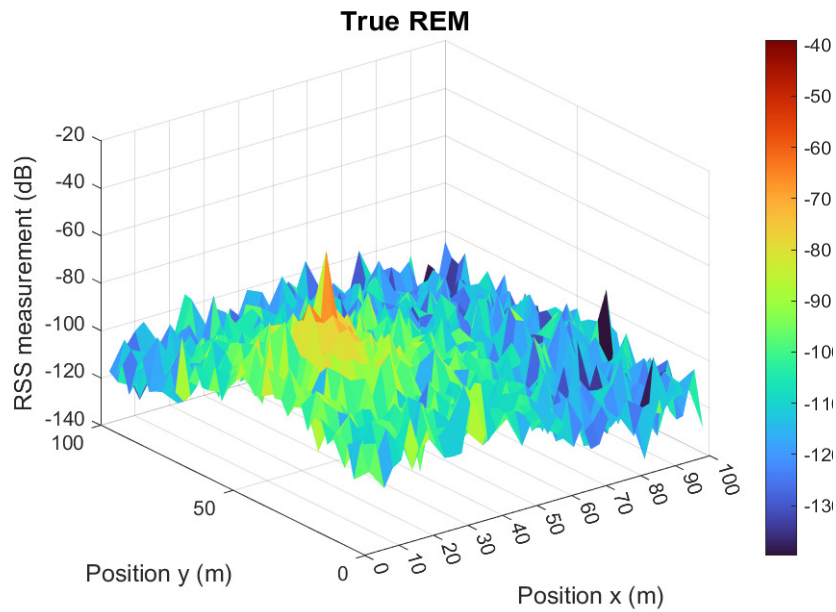
Figure 5.5: True and estimated REM without additive noise.

The two figures 5.7 (a) and (b) show the impact of log-normal shadowing noise on the power maps. The RSS variations are clearly evident in the yellow and green zones, with reduced blue regions of lower RSS measurements within the area of interest. The two figures are mapped

with 40×40 samples containing the RSS measurements at points where the RSS measurements were not measured or available. We employ the use of interpolation to map the power levels for those sampled areas in the grid.

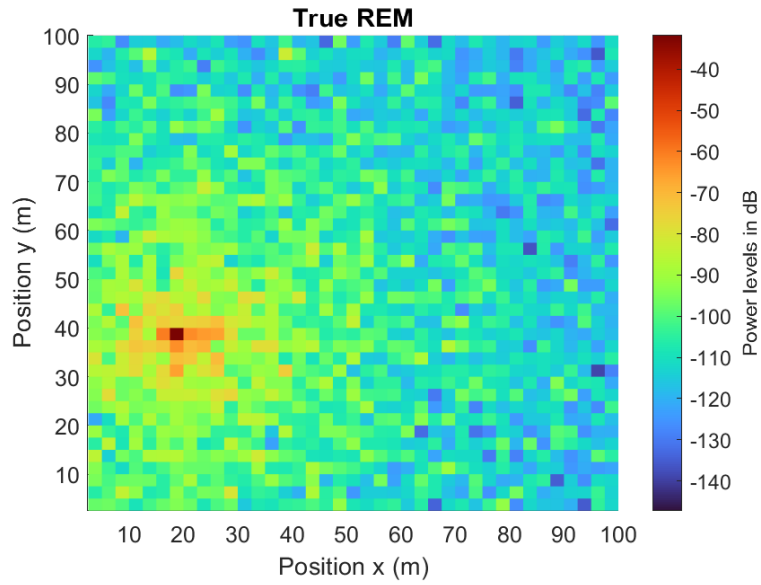


(a) 3D view of true REM.

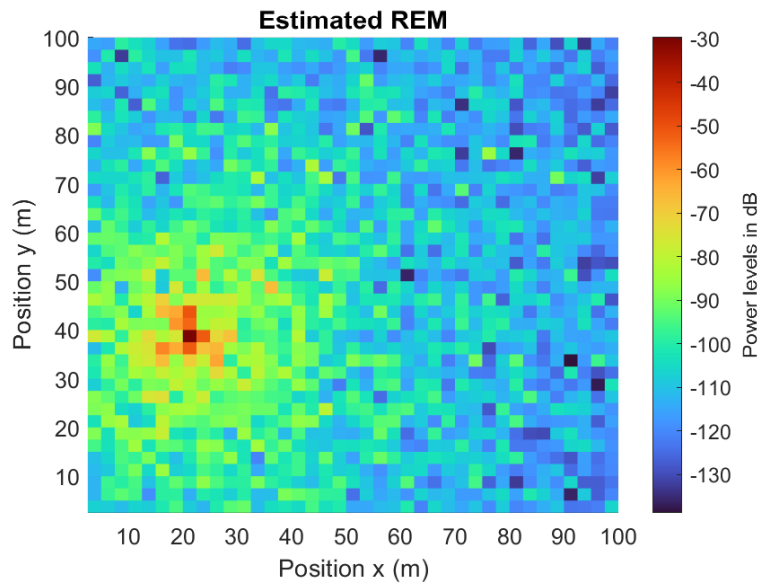


(b) 3D view of true REM with added noise.

Figure 5.6: 3D view of the REM.



(a) True REM with noise.



(b) Reconstructed REM with noise.

Figure 5.7: True and reconstructed REMs with log-normal shadowing.

5.3 Extension of system model

In the complementary section, we address extension to the aforementioned system design, including transmitter signal waveform generation (OFDM signal) and modeling correlated shadowing noise as opposed to log-normal independent shadowing as in the initial system model. The extensions are essentially practical and relevant for real-world scenarios and look at how they impact the performance of localization methods in general.

In the following sections, we first address the situation with the use of OFDM waveform as a transmit signal rather than the use of static transmit power. We delve into OFDM structure and its basic concepts and work our way to simulate the 5G NR waveform and infer the localization performance. Further, we model our shadowing as a correlated model as we witness the impact of correlated links between sensor nodes employing the use of an appropriate shadowing model and compare the performance of the localization methods from earlier.

5.3.1 OFDM waveform-based transmitter localization

In this section, we aim to exploit digital signal processing for generating signal waveforms such as OFDM, radar beacons, etc., for the purpose of sensing. In order to provide relevance for 5G networks, we use OFDM symbols to carry out localization. The system parameters with ultra wide bandwidth and frequency of operation in the sub 6 GHz range render other channel parameters the same as before. Our scenario does not include the mobility of sensors or even UE transmitters. We focus entirely on static scenarios and avoid complications regarding Doppler shift and beamforming employing multiple input and output systems. First, we start with the basics of OFDM and its structure, introduce the 5G NR radio frame structure and system model, and present our findings for localization.

5.3.1.1 OFDM basics

OFDM is a combination of modulation and multiplexing. Multiplexing suggests combining signals from various sources, such as antenna components together and modulation is impressing data as digital to improve spectrum usage and quantize data to reduce space consumption. OFDM is essentially orthogonality in the frequency domain i.e., maintaining orthogonality between sub-carriers in a bandwidth. The main principle for OFDM is the orthogonality between sub-carriers, essentially the Fourier transform's multiplicative factors regarding sine and cosine waves. The orthogonality is explained as in a time period T_s of one symbol duration the area under the curve of sine and cosine waves is zero. Hence, the orthogonality. The exact sense of orthogonality of OFDM waveform is described using the following figures.

As stated above, figures 5.8 (a) and (b) sufficiently explain the orthogonality between the subcarriers. The interference from the other subcarriers is essentially nullified since the peak value of a particular carrier is at the time instant where the other signal value is zero. Figure 5.8 (a) shows the time and frequency view of one OFDM symbol with N sub-carriers. As it is clear from the figure, different subcarriers have different carrier frequencies and amplitude in the allocated symbol duration T_{symbol} . One OFDM symbol for a symbol duration is calculated as the sum of all the orthogonal subcarriers. The exact working system model and functional diagram of one OFDM symbol are discussed next.

The functional block diagram for the OFDM signal generation is described in figure 5.9. The essential parts include the modulation of bit sequence and encoding, then through serial to parallel converter feeding the sequence in each time step to IFFT block summed up together and sent further through the channel. The receiver side performs the opposite operation first downsampling the received signal, removing the cyclic prefix, then passing it through the FFT block, then demodulating, and lastly passing it through parallel to the serial converter.

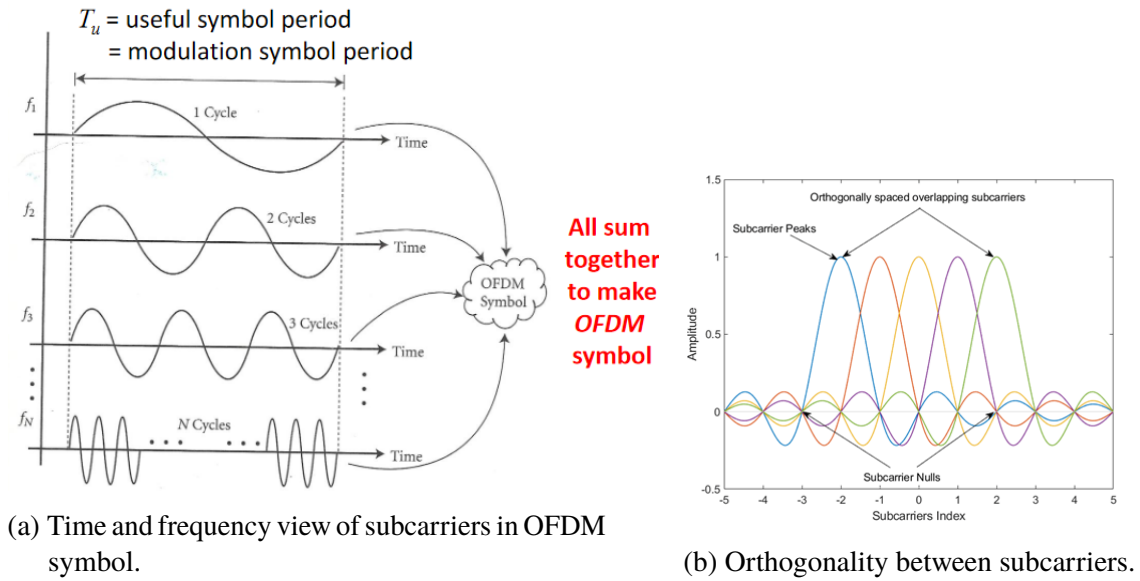


Figure 5.8: Basic view of carrier frequencies of OFDM and orthogonality between different sub-carriers [55].

The size of the FFT and IFFT blocks is given as N , which is also the number of subcarriers, and each of these subcarriers are separated in the frequency domain by a spacing of $n\Delta f$. The spacing is integer multiples of Δf , which maintains the orthogonality between the subcarriers. The IFFT block inputs the N complex signal in the frequency domain and gives out the output in the time domain. The mathematical formulation for IFFT samples is given by:

$$X(n) = \frac{1}{N} \left(\sum_{k=0}^{N-1} x(k) \sin \frac{2\pi nk}{N} - j \sum_{k=0}^{N-1} x(k) \cos \frac{2\pi nk}{N} \right) \quad (5.2)$$

Where n is the subcarrier index and k is the complex coefficient for each frequency bin sample. These time domain samples are upsampled and transmitted over the channel, and the receiver performs the inverse steps to return to the original sequence. The mathematical formulation for the FFT sequence is given by:

$$x(k) = \frac{1}{N} \left(\sum_{n=0}^{N-1} x(n) \sin \frac{2\pi nk}{N} + j \sum_{n=0}^{N-1} x(n) \cos \frac{2\pi nk}{N} \right) \quad (5.3)$$

$x(k)$ is the frequency bin samples as the output of the FFT process. A simplified block diagram and signal processing structure are shown in figure 5.10. It simplifies the complex processes involved in OFDM symbol generation using the most important functional blocks. In our system model, we employ the use of such simpler functionality rather than producing a more involved and complex process.

A simpler and more precise view of all the involved concepts for OFDM symbol generation can be found in figure 5.11. It explains the time and frequency view of the subcarriers and a

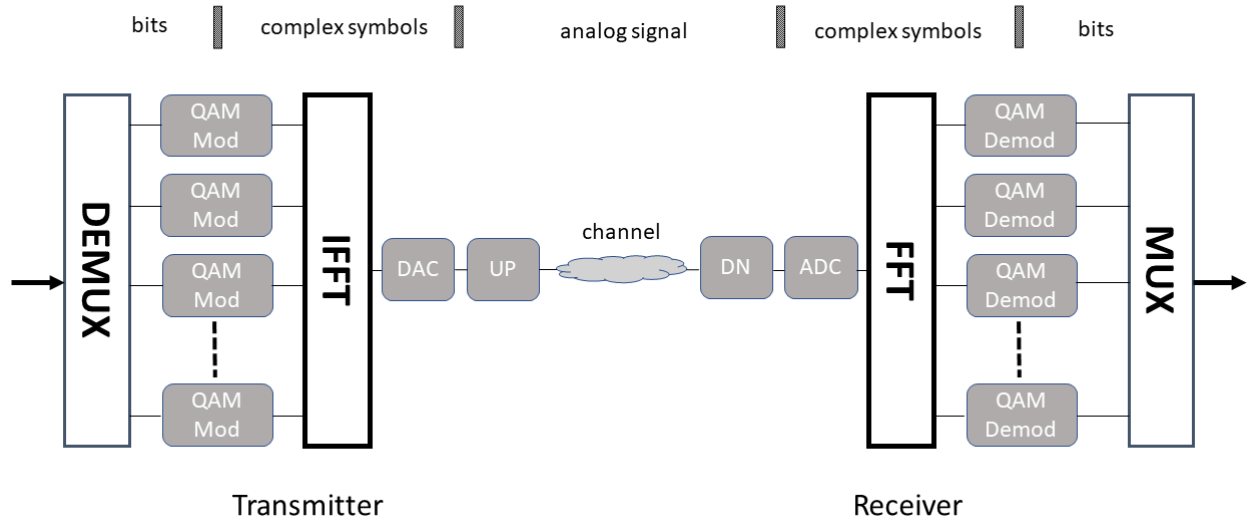


Figure 5.9: Functional OFDM block diagram.

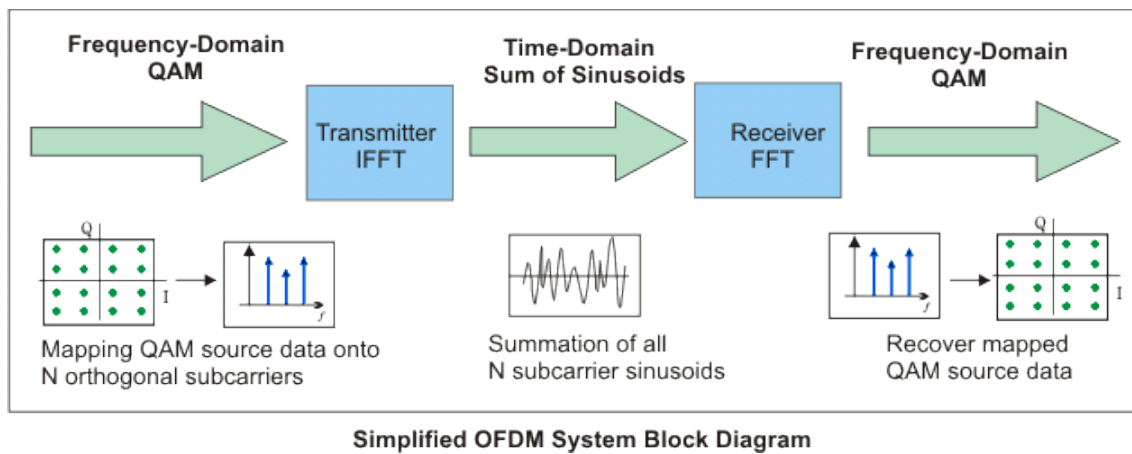


Figure 5.10: Simplified OFDM symbol generation [56].

summed-up signal over all the subcarriers over the frequency domain for each QAM sequence for a time period T_s seconds.

Another advantage of using OFDM is its robustness to interference from other subcarriers. As shown in figure 5.12, a small segment or some part of the last occurring samples of an OFDM symbol are added as a prefix to the same signal, which makes it robust to interference, and this cyclic prefix part is removed before performing FFT operation at the receiver side.

Some limitations of OFDM signal are:

1. High peak to average power ratio (PAPR).
2. Inter-carrier interference.
3. Carrier frequency offset.

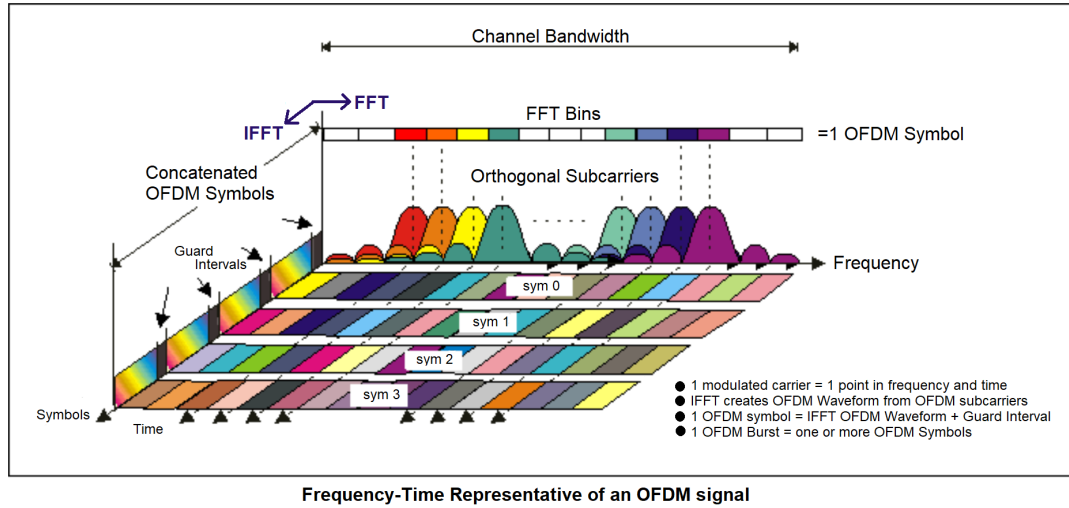


Figure 5.11: Time and frequency view with additional orthogonality view of frequency bins in an OFDM symbol [57].

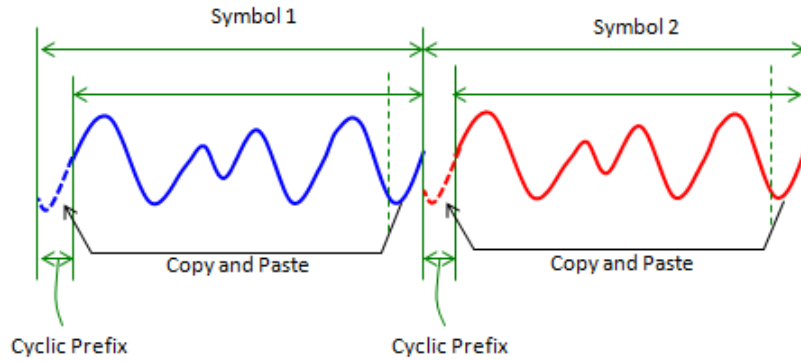


Figure 5.12: Cyclic Prefix.

5.3.1.2 5G NR frame structure

After discussing some basic OFDM terminology, we dive into the 5G NR radio frame numerology, which is quite relevant to our case of performing localization using OFDM signal as a means to replicate the 5G NR waveforms. The 5G NR frame structure is more adaptable than its predecessors, such as LTE, as it is more application-based rather rigid for most applications e.g., IoT, autonomous vehicles, etc. One key aspect of the frame structure is its resource blocks, including time-frequency blocks spread across the physical resource blocks (PRB). The whole resource block is divided into several resource elements, each containing QAM or modulated symbols. This ensures fast and optimal distribution of information. The following figure gives us insight into the PRBs, resource elements, and simple frame structure of one OFDM symbol.

As depicted in the figures 5.13, the 5G NR radio frame is split into two simpler components called subframes and slots. The N_{symb}^{slot} terminology is quite important to understand the structure of the radio frame and its components, it explains the underlying divisions of a single radio frame and subframes into N symbols in a slot. To get a more detailed view of the frame division we refer to Fig 4.14 (b), where a single frame is split into two half frames which in turn consist

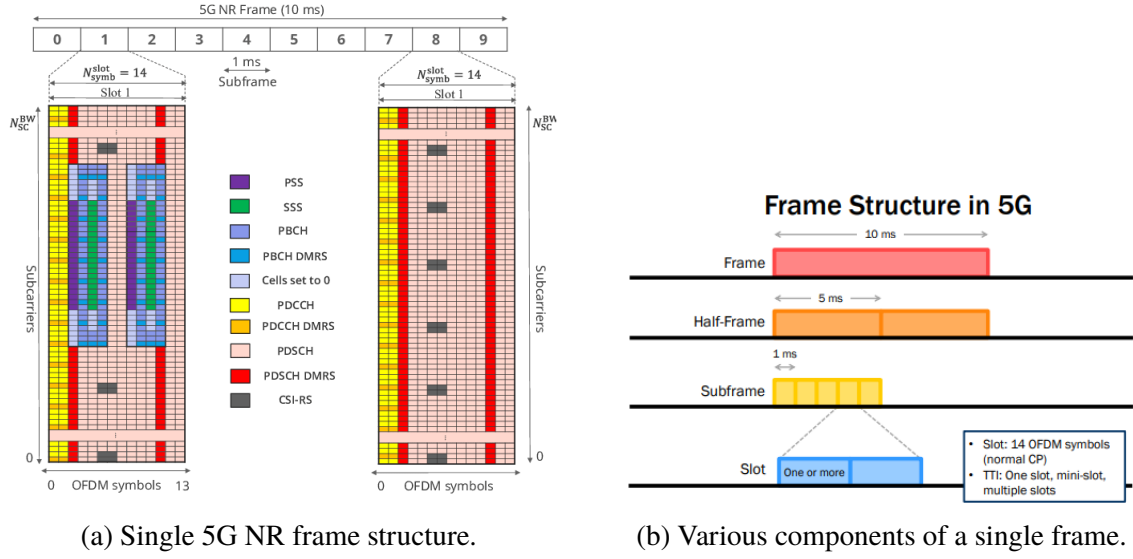


Figure 5.13: 5G NR frame structure and its components [58].

of several subframes, and then each subframe could be split into slots. The coloured elements in the resource grid represent the signal types such as PDSCH, PUSCH, SRS PDCCH, etc. which are out of the scope of the thesis and we focus on random data generation and assignment to the OFDM symbols.

To completely understand the frame numerology concept it's better to consider an example of numerology $\mu = 0$. The time period of a single frame is defined as $T_f = 10\text{ms}$. For numerology 0 the frame is divided into 10 subframes and each subframe is of length $T_{subframe} = 1\text{ms}$. Again each subframe is divided into slots, and in numerology 0, each subframe consists of 1 slot. In conclusion, a frame contains 10 subframes and 10 slots each. Also, each subframe contains 14 OFDM symbols which are spread out in the resource grid as resource elements and one OFDM symbol's length is $66.67 \mu\text{s}$ which translates to the subcarrier spacing δf is 15 KHz.

A detailed overview of the numerology terminology as a table is described in Appendix 1 for further reading.

5.3.1.3 OFDM based system model

Figure 5.14 shows the complete and precise sample setup for generating OFDM waveform with system parameters. The pilot symbols for each OFDM are placed at subcarrier positions as shown in the figure. The input bitstream is fed to the encoder and modulation block and then to serial to a parallel converter then, we add a cyclic prefix and feed it into the OFDM generator function in MATLAB. For our case, we use system parameters such as constellation size $M = 4$ for quadrature amplitude modulation (QAM), number of subcarriers $N = 128$, cyclic prefix length 16 symbols samples, total number of OFDM symbols 28 for an observation window of 2 ms. We combine the signal strength of 28 OFDM symbols to improve the SNR of the signal used for localization. The system parameters remain the same from section 5.1.

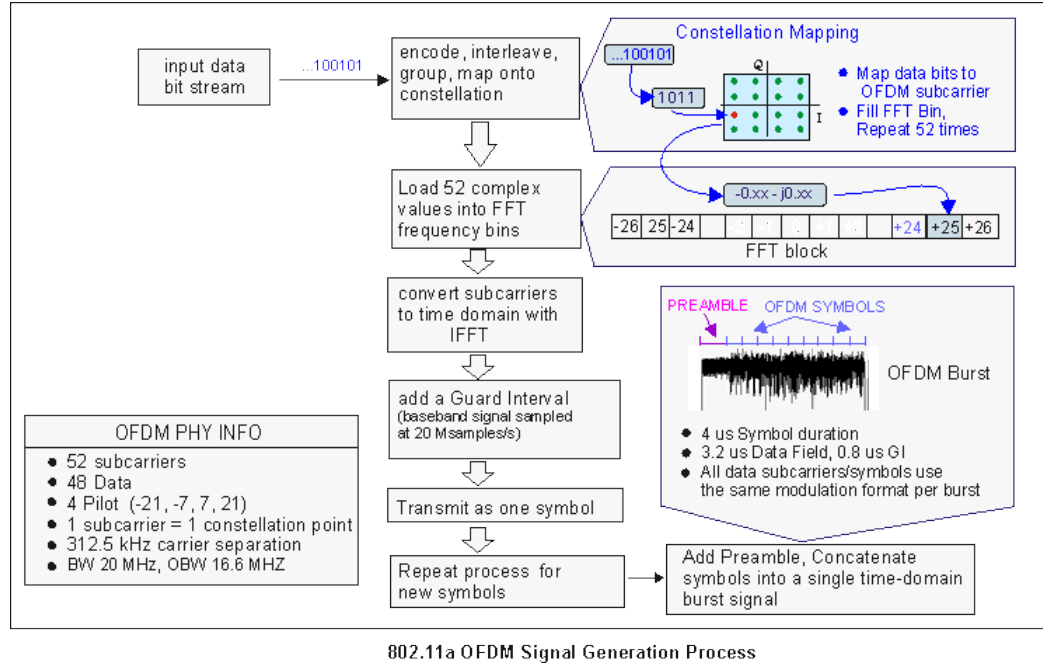


Figure 5.14: System model for OFDM waveform generation [56].

5.3.1.4 Simulation results

The simulation results for the OFDM signal-based transmitter localization are presented in this section. We first provide the simulation for generating OFDM waveform based on system parameters from section 5.3.1.3. The OFDM waveform depicted in the plot contains 4032 samples, constituting the 28 OFDM symbols [59] with corresponding time window as compared to 5G NR radio frame with numerology 0 being 2 ms, each consisting of 144 samples (128 FFT size + 16 cyclic prefix length). The frequency of operation is chosen to be 3.7 GHz [60].

In the following figures, we compare the localization performance of the MSE-based approach with the OFDM waveform. We present the result with system parameters the same as in section 5.1, and instead of static transmit power from an unknown transmitter, we use the OFDM waveform as shown in figure 5.15. The OFDM waveform as a transmitted signal performs better in localization tasks than fixed transmit power, as presented in section 5.1.

Figure 5.16 shows that using OFDM signal, the MSE approach performs better than the scenario with fixed transmit power with localization error less than 1 m or around centimeter-level precision as the number of deployed sensors reaches 9. This turns out to be a very interesting result, as OFDM symbols provide quite accurate results while performing localization even though the localization accuracy employing RSS measurements is usually in the range of 1 m to 10 m, apparently [5]. Part of the problem statement was to employ the use of OFDM signal as a means to employ real-world signals instead of simple transmit power for analysis, but the true reasoning behind the outcome, as shown in figure 5.16, is beyond the scope of the thesis.

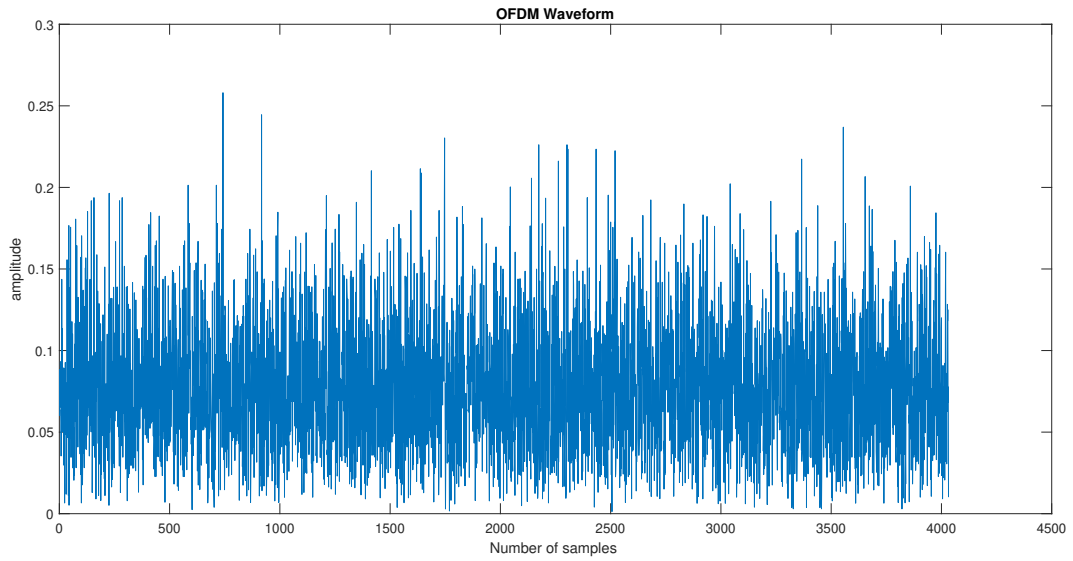


Figure 5.15: OFDM waveform.

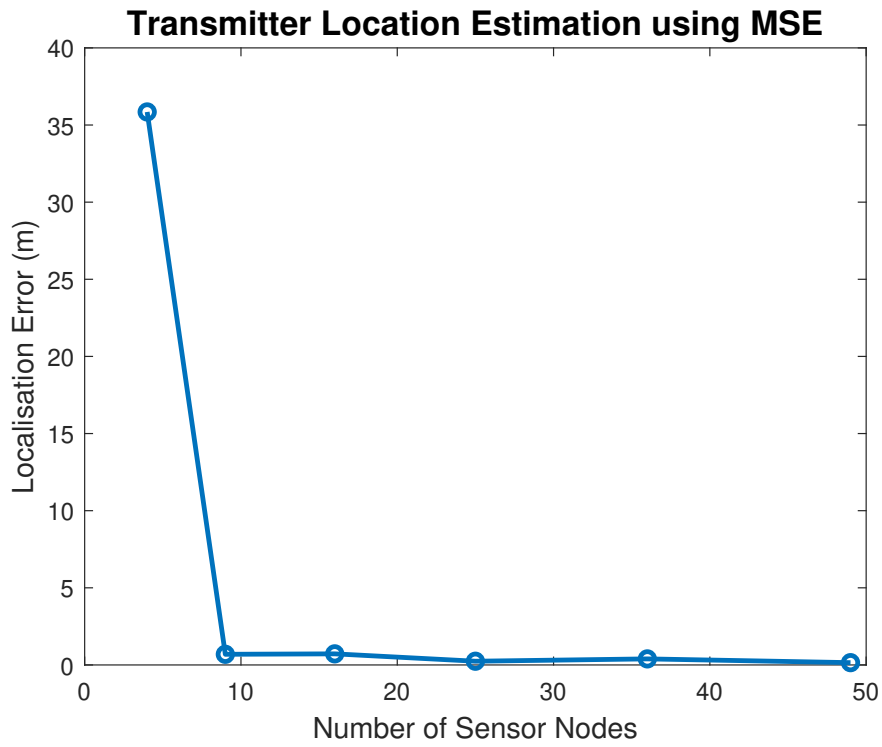


Figure 5.16: Localization error using OFDM signal for grid of size (100mx100m).

5.3.2 Correlated shadowing

Another important aspect to take into account when studying sensor network sampling is the correlation among received power measurements taken at relatively short distances from each other. This correlation arises because signals undergo similar propagation effects (i.e.,

blockages, reflections, and scattering) when the measuring receivers are located near each other. The literature generally considers two fading strategies: first, fading is insignificant, meaning the links in the wireless channels are free of fading losses, or fading doesn't change the attenuation for a transmitter-receiver pair. Another strategy is that fading is independent or statistically independently (i.i.d.) distributed. This i.i.d fading considers the fact that every link is independent and independently affects the receiver. However, in a scenario where the links to the receiver are close enough are also affected in a similar fashion by the neighbouring shadowing objects. This behaviour forces us to think and consider the importance of correlated shadowing losses in nearby links as such is the case in real-world scenarios. The idea behind employing correlated shadowing is not to be confused with the increase in performance of the localization method but to demonstrate the scenarios as they are applicable in real world cases. Therefore, the main idea behind this is choosing and employing the use of an appropriate correlated shadowing model in a physical environment. In general correlated shadowing is related to the environmental changes caused by the mobility of the transmitter or the receivers, in other words, Doppler shift. However, in our case, we employ just the simpler and adaptable scenario where we model correlations between the standalone links between transmitter and receiver structure.

In our previous discussion in section 5.1, we used the i.i.d log-normal shadowing model, which is extensively used in the literature for localization purposes. This section extends our system model to the Gudmundson model for correlated shadowing. The shadowing distribution is given by $\mathbf{w} = [w_1, w_2, \dots, w_N]$ where $\mathbf{W}$ is normally distributed as $\mathbf{w} \sim \mathcal{N}(\mu, \Sigma)$, where the dimension of mean and co-variance matrix is $\mu \in \mathcal{C}^{N \times 1}$ and $\Sigma \in \mathcal{C}^{N \times N}$ respectively. The co-variance is defined using the Gudmundson model:

$$\sum_{i,j} = \sigma^2 \exp\left(\frac{-|u_i - u_j|}{d_{corr}}\right) \quad (5.4)$$

where u_i is the i -th sensor node, d_{corr} is the decorrelation distance and σ is the standard deviation.

The authors in [28] propose a particle simulation-based method for localization using correlated shadowing. The authors consider particle simulation and correlated shadowing to account for mobility and varying Doppler. They compare the performance of the localization method based on various standard deviations of the noise random process. They conclude that as the shadowing between sensor nodes is more correlated, the localization method performs worse, and the accuracy decreases as compared to the process of no shadowing i.e. $\sigma_S = 0$. In [61], the authors propose a differential RSS-based localization method and compare the performance of the localization method for two cases: a) Log-normal shadowing, and b) Correlated shadowing. The differential RSS-based method performs better as the correlation between sensor nodes increases or is present than the case with mismatched RSS measurements. In the case of almost uncorrelated showing (normalized covariance coefficient $\sigma_S = 0.4$), the differential RSS approaches the RSS method regarding localization accuracy. However, as the correlation increases, there is a change in localization accuracy.

5.3.2.1 Simulation results

After choosing an appropriate system model design, we plot the simulation results comparing the performance of the localization using MSE-based approach for varying shadowing standard deviations. The system parameters used for this comparison are the same as mentioned in section 5.1. We select standard deviation of the system $\sigma_S = [0, 4, 8]$ dB, and the decorrelation distance $d_{corr} = 8$ m [13].

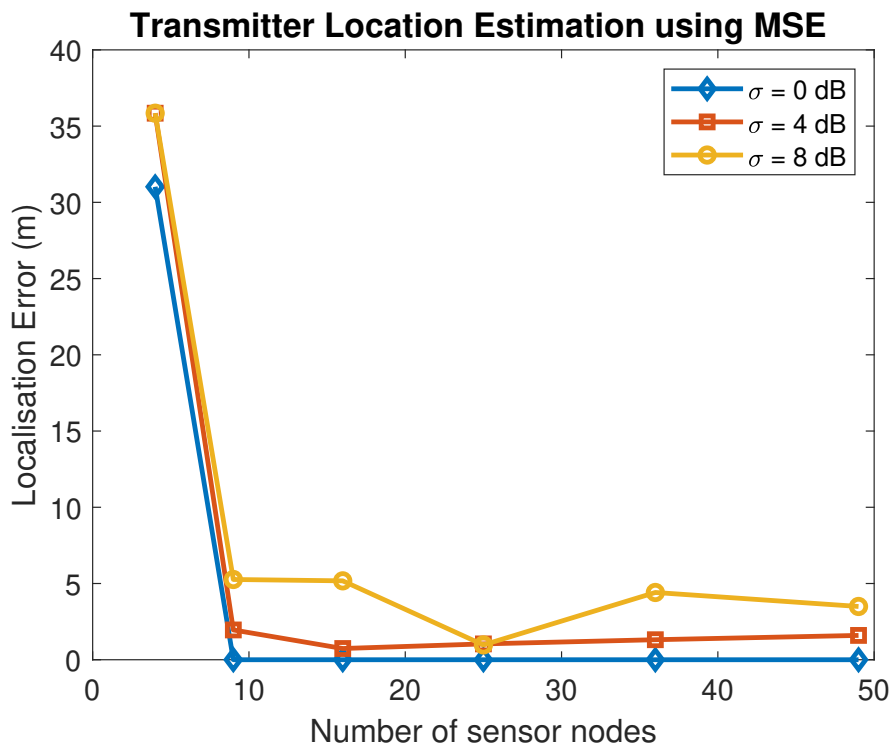


Figure 5.17: Performance of MSE-based localization method for varying standard deviation coefficients.

In the figure 5.17, as we increase the standard deviation, we incorporate more shadowing losses from the surrounding environment which means more correlation amongst the neighboring sensor nodes and less correlation as the separation between the sensors increases. This inference comes as the increase in standard deviation causes an increase in shadowing correlations between the sensor links, and the sensor links are treated as similar, which may result in similar RSSI levels, thus employing no improvement in localization even though the number of deployed sensors increases. Another aspect to note is the decorrelation distance and the shadowing model's standard deviation. From the literature, the decorrelation distance of the indoor scenario, in general, is considered between 5 -10 meters, and we choose the value in between the extremities of the range as 8 m. The higher the standard deviation, the more losses incurred in the localization method.

5.4 Chapter summary

In the following chapter, we compare the performance of the two localization methods, namely MSE and L_{IV}E. Both of the localization methods perform similarly in terms of localization accuracy. However, MSE method incurs lower run time complexity, and we choose MSE approach for REM construction. Additionally, we extend the defined system model to incorporate 5G NR waveform and correlated shadowing and analyze their effect on classical localization methods for immobile scenarios.

6 Conclusion

6.1 Conclusion

The main contributions of the thesis are, firstly, evaluating the two localization methods described in the literature. From the results in section 5.1.1, we conclude that MSE-based approach localizes the transmitter as close to the true position with localization error under 5 meters in the area of interest. However, the trend of increasing localization error is seen as we increase the area of interest which spans (100m, 100m). Therefore, making it essential for RSS-based localization to place sensors close to the transmitter and not very far apart. The results from section 5.2 indicate that LlvE performs better than MSE in terms of localization error with an accuracy of localization close to 1 m. The comparison in plot 5.4 shows the error difference is high when the number of sensors is 4, but the performance of the two methods converges as the number of sensors increases from 16 to 49, and the difference of localization error between the two methods is in the range of 1 to 3 m approximately. The evaluation of the two localization methods helps us to choose the MSE method as a favorable method in terms of trade-off between localization accuracy and better simulation time in MATLAB as stated in section 5.2. A limitation, as mentioned in section 5.1.2 is the infeasible point convergence of the LlvE method as the optimization problem converges to an infeasible point and needs to be defined more clearly to satisfy the non-linear constraints.

The thesis also evaluates the use of OFDM waveform as an alternative transmit signal (using signal processing) instead of static transmit power, as is used widely in literature, as shown in figure 5.16. The exact reason for the shown performance in section 5.3.1.4 is unclear to the author because of time constraints, and the produced result was unexpected. Another aspect of evaluating the localization accuracy on the RSS-based localization is using a correlated shadowing model such as Gudmundson model as explained in section 5.3.2, the standard model to incorporate correlations in a dedicated WSN. As the standard deviation of shadowing increases, the localization error using MSE method increases by a margin of around 2 m for averaging over 25 experiments for varying standard deviations $\sigma_S = [0,4,8]$ dB. This particular result explains how correlation between the sensor node links is important to consider for localization as the environment for WSN is highly dependent on shadowing.

The result from the localization methods, the MSE approach, is then used to construct REM after localization as described in figures 5.5 and 5.7. The resultant power maps helps in locating the presence of active transmitter in the region of the highest power levels. A high similarity was found in the true REM and estimated REM after using results from the localization method to construct the RSSI levels in the given area of interest. We further discuss of relevance and importance of REM as a technology not only for CR but also for 5G based applications and use cases such as high speed trains, V2X based applications to inform the central server about

the vehicle's route and the environment to avoid accidents. REM also can be used to support the deployment of virtual base stations through UAVs in areas where connectivity is severely hindered.

6.2 Future work

The current scenario for this particular work was considering slow fading, however as the present technologies work in high frequencies, and fast-changing fading environments, there is a need to extend the localization using WSN for fast-fading cases, which are more relevant in mmWave and mobile scenarios such V2X. From the results there is room for improvement in MSE and LlvE methods regarding localization accuracy and conforming to fast fading. The simulation evaluation employing the use of OFDM and its corresponding performance in large-scale and small-scale fading is left for further research as it was not the explicit goal of the thesis, and the exact justification for the results in section 5.3.1.4 is unknown due to time constraints. REM construction was only considered in 2D scenarios but another extension to REM construction methodology could be incorporating spatial dimension resulting in 3D scenarios. As the thesis considered evaluation localization in static scenario, an important use case is to implement and model the dedicated sensor network in mobile scenarios which implicitly adds the Doppler shift and correlations between the dedicated sensors and transmitters.

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2.1 Comparison of Centralized and Distributed Spectrum Sensing [10]. 8

Appendix

1 Appendix 1

The following table presents the actual numerology used for defining the radio frame structure for 5G NR.

μ	$N_{\text{ymb}}^{\text{slot}}$	$N_{\text{frame},\mu}^{\text{slot}}$	$N_{\text{subframe},\mu}^{\text{slot}}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16
5	14	320	32
6	14	640	64

2 Appendix 2

Radio spectrum is a natural resource that is heavily utilized by dense networks comprised of many users, which burden the allocated bandwidth and thus cause performance degradation in capacity and data consumption. As a result, FCC in America frees up 150 MHz bandwidth in the frequency range of 3.5-3.7 GHz allocated for private networks in 4G LTE specifically for military use [62]. The CBRS band consists of 150MHz within the 3.5 GHz spectrum band that stretches between the 3550-3700MHz (or 3.55-3.7GHz). Hence, citizen broadband radio services (CBRS) as a solution to use the free bandwidth to accommodate users or use cases in private networks to ease the burden on public networks and still provide high quality data usage. This was introduced as a solution in LTE but CBRS also provides its services in 5G to suffice for resource allocation such as bit rate, capacity and high volume of users. It is highly likely that the market for LTE and cellular systems will undergo a significant transformation with the advent of private LTE networks that are as easy to access as WiFi networks and available free of cost. As a result, we can expect a shift from the current scenario of only a few nation-wide LTE networks to the emergence of numerous small-scale LTE networks. [60].

2.1 The CBRS framework

The CBRS spectrum sharing rules were created to provide wireless access to the public while also safeguarding incumbent users from interference. These rules are critical because the United States Military and Navy already use bands 42 and 43. The CBRS framework consists of three tiers arranged hierarchically. This model creates new opportunities for the distribution and use of wireless access. The first tier is where Incumbent systems are placed, such as Navy RADARs, US Military, and Satellite Stations. This tier has priority over 150 MHz, but it may only use some of the band at some locations. The second tier prioritises access devices, referred to as the Priority Access License (PAL). This tier offers a non-renewable 3-year license paid to the FCC for using a 10 MHz channel within the 3500-3650 MHz portion of the band, limited to a specific geographic area. The PAL license takes priority over Tier 3. Finally, the lowest tier (Tier 3) is called the General Authorized Access (GAA). The GAA is a dynamic allocation of 100 MHz channels within the entire band (3500-3700 MHz).

2.2 Uses of LTE band 48- CBRS

Band 48 - CBRS: Band 48 CBRS will likely be used for private and geographically restricted LTE deployments, which is superior to WiFi. Mobile broadband internet providers (PALs) can use Band 48 to deploy Public LTE Networks and improve their service by extending it. To accomplish this, they will need to install LTE hardware for Band 48, such as indoor and outdoor antennas, and wireless gateways. Other PALs will likely use private LTE for industrial IoT applications. CBRS can be easily supported by 5G devices that support it, such as the Apple iPhone 11, which could be used for applications like localization, security, and other 5G use cases [62].